

Learning and Free-Riding in International Climate Policymaking*

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September 24, 2026

Abstract

International climate cooperation is often presented as a case of the tragedy of the commons, where individual countries have incentives to free-ride on other nations' emissions reduction efforts. I present empirical evidence to the contrary: countries' climate actions are positively correlated over time, exhibiting a complementary pattern across nations. I explain this discrepancy with a formal model in which other countries' mitigation efforts signal information about the potential economic benefits of decarbonization in addition to providing the global public good of reducing emissions. The model identifies when these informational effects outweigh free-riding incentives and produce complementary policymaking. Empirically, I show that governments increasingly cite foreign climate policies when evaluating the domestic economic benefits of mitigation. Using European public opinion and cross-national policy data, I also find that foreign climate action is associated with more favorable beliefs about climate policy and that these beliefs predict subsequent mitigation law adoption.

*Many thanks to Tara Slough, Alastair Smith, Sandy Gordon, Peter Rosendorff, Pat Egan, Bruce Bueno de Mesquita, Amanda Kennard, Matto Mildenberger, Carolina Torreblanca, Aleksandra Conevska, Rajeshwari Majumdar, Joe Ruggiero, Margaret Kenney, Nitheesha Nakka, and audiences at New York University, Harvard University, University of Texas at Austin, MIT, University of Wisconsin-Madison, PECE, and APSA for thoughtful comments and feedback.

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How does the behavior of other nations influence interstate climate policy adoption? Long-standing scholarship emphasizes the temptations to free-ride inherent to collective action concerns (e.g., [Olson 1965](#); [Ostrom 1990](#); [Barrett 2003](#); [Stern 2007](#); [Bernauer 2013](#); [Keohane and Victor 2016](#)). Per this view, we should observe effort substitution: the costs of mitigation are borne locally while its benefits are shared globally, and there are diminishing marginal returns to global abatement. Together, this implies that one country’s mitigation efforts reduce the incentives for others to do the same.

This paper argues that free-riding is not the only consequence of foreign climate action: mitigation efforts pursued abroad also allow governments to learn about the potential economic returns to decarbonization. Nations face uncertainty about whether green investment will generate domestic gains—including new industries, jobs, lower energy costs, or access to emerging supply chains. Observing foreign policy experiences provides information about these possibilities, offering an additional explanation for the spread of large-scale mitigation efforts that free-rider effects struggle to explain alone. Cross-border learning is especially consequential in this context because international environmental agreements remain weak and no country can unilaterally supply the public good of mitigation. I thus contend that foreign climate policies have two consequences for other governments: they reduce incentives for additional mitigation by increasing the provision of a global public good, but they also reveal information about the economic value of a green transition.

To examine how these two effects jointly shape climate policymaking, I develop a formal model of public goods provision with interstate learning. Upon observing mitigation efforts abroad, countries continue to face the same free-rider incentives emphasized by existing theories, but also update their beliefs about the potential economic returns to decarbonization. The model characterizes how these two effects interact to determine nations’ climate policy choices and identifies the conditions under which learning outweighs free-riding, producing complementary rather than substitutable climate policies across countries. The central

implication is that the tragedy of the commons is not an inevitable consequence of international climate politics. Rather, it is an equilibrium outcome that depends on whether the informational benefits generated by foreign climate action can offset free-riding temptations.

I present the argument in three steps. I first document that climate policies are positively correlated across countries using several measures of climate ambition. These patterns are difficult to reconcile with effort substitution alone, and suggest that free-riding may not be the only mechanism affecting international climate policymaking. To explain this empirical finding, I then develop the formal model—positing cross-border learning as an explanation for effort complementarities. Finally, I evaluate the proposed learning mechanism in two ways. I show that governments increasingly justify climate policy in terms of its domestic economic benefits based on other nations’ experiences through a corpus of cross-border citations within national climate documents. I then derive and test several empirical implications of the theory, combining European public opinion data with cross-national evidence on climate policy adoption.

Scholarship on international climate cooperation is primarily rooted in theories of collective action and public goods provision. Because the benefits of emissions reductions are shared globally while the costs are borne locally, governments have incentives to free-ride on the efforts of others ([Ostrom 1990](#); [Dietz, Ostrom and Stern 2003](#); [Victor 2011](#); [Nordhaus 2015](#); [Busby and Urpelainen 2020](#)). This logic underpins a wide range of scholarship on climate cooperation, from models of treaty formation and participation to analyses of institutional design and compliance (e.g., [Barrett and Stavins 2003](#); [Calvo and Rubio 2013](#); [Hovi, Ward and Grundig 2015](#); [Harrison and Lagunoff 2017](#); [McAllister and Schnakenberg 2022](#); [Hai 2026](#)). Across these accounts, other countries’ mitigation efforts influence climate policymaking because they alter the strategic incentives facing governments by changing the provision of the global public good. This perspective has generated fundamental insights into the challenges of international climate cooperation; I complement this literature by showing

how climate policies themselves generate informational spillovers that promote subsequent policy adoption, even in institutional settings that provide relatively weak incentives for cooperation.

This paper augments standard accounts of international climate cooperation by incorporating insights from the policy diffusion literature. Research on international climate politics has overwhelmingly emphasized how foreign climate remediation diminishes incentives for further mitigation through the logic of collective action ([Barrett 2003](#); [Hovi, Ward and Grundig 2015](#); [Harrison and Lagunoff 2017](#); [Weitzman 2017](#); [McAllister and Schnakenberg 2022](#); [Kennard and Schnakenberg 2023](#)). The policy diffusion literature offers a complementary perspective—governments routinely observe one another’s policy experiences and learn from them ([Elkins and Simmons 2005](#); [Simmons, Dobbin and Garrett 2006](#); [Volden, Ting and Carpenter 2008](#); [Gilardi 2012](#); [Carattini, Gosnell and Tavoni 2020](#)). I argue that synthesizing these perspectives yields a richer account of international climate cooperation: foreign climate policies influence other governments not only by lessening the marginal value of their own mitigation, but also by revealing information about the economic returns to decarbonization. Climate policies may thus become complementary rather than substitutable across countries, generating policy diffusion even in the presence of persistent incentives to free-ride.

The policy diffusion literature demonstrates that policy adoption conveys information, as governments learn from one another’s policy experiences, updating their beliefs about policy effectiveness after observing implementation abroad ([Elkins and Simmons 2005](#); [Simmons, Dobbin and Garrett 2006](#); [Volden, Ting and Carpenter 2008](#); [Gilardi 2012](#)). Evidence of such learning has been documented across a wide range of policy domains, including environmental policy, where scholars have identified cross-border diffusion in carbon pricing ([Harrison 2010](#); [Ward and Cao 2012](#); [Thisted and Thisted 2020](#); [Linsenmeier, Mohommad and Schwerhoff 2023](#)), feed-in tariffs ([Baldwin, Carley and Nicholson-Crotty 2019](#)), and domestic and

international climate legislation ([Bernauer et al. 2010](#); [Sauquet 2014](#); [Fankhauser, Gennaioli and Collins 2016](#)). Recent work also finds “ratcheting up” effects based on diffusion in countries’ climate pledges ([Rowan 2025](#)). This paper extends these insights, arguing that governments update their expectations about the economic returns to climate remediation by observing mitigation policies adopted abroad, and does so in the context of free-riding temptations.

Governments contemplating ambitious climate policies face substantial uncertainty about their domestic economic consequences. Although climate policy has traditionally been understood as imposing local costs in exchange for global environmental benefits, a growing literature argues that mitigation may also generate important domestic benefits, including technological innovation, the growth of nascent green industries, and broader economic development ([Porter and van der Linde 1995](#); [Rodrik 2014](#); [Keohane and Victor 2016](#); [Cullenward and Victor 2020](#); [Hepburn et al. 2020](#); [Sabel and Victor 2022](#)). As the empirical evidence in this paper shows, decarbonization has become increasingly justified not only as environmental protection, but also as a strategy for economic development; and these justifications are often based on the policy experiences of other nations.

Recognizing that decarbonization may generate substantial domestic economic benefits has also reshaped scholarship on the politics of climate policy. A burgeoning literature argues for a focus on domestic distributive cleavages to explain variation in climate policymaking (e.g., [Aklin and Mildenberger 2020](#); [Colgan, Green and Hale 2021](#); [Ross 2025](#); [Gazmararian, Mildenberger and Tingley 2026](#); [Gazmararian and Tingley 2026](#)). In these accounts, expectations about the benefits and costs of decarbonization underpin the adoption of climate reforms, with some claiming that international collective action is irrelevant for explaining mitigation ([Urpelainen and Van de Graaf 2018](#)).¹ I provide an international rationale for the

¹In eschewing free-riding, scholars of redistributive politics sidestep the question of how international factors affect climate policy decisionmaking—heterogeneity of preferences domestically could still be compatible with a free-riding story at the international level ([Kennard and Schnakenberg 2023](#)).

domestic distributional dynamics highlighted within this literature: policy diffusion can be a source of learning about the benefits of climate remediation, which then animates domestic political conversations about winners and losers.

How Do Countries' Climate Actions Correlate?

Standard accounts of global climate policymaking posit that the marginal value of action declines as other countries do more, since benefits are shared globally while costs are borne locally. This logic implies incentives to free ride and is often formalized by assuming that countries' actions are strategic substitutes (see [Harrison and Lagunoff 2017](#); [McAllister and Schnakenberg 2022](#); [Kennard and Schnakenberg 2023](#)).² Under this assumption, greater mitigation by others reduces a country's optimal efforts, generating a negative correlation in cross-national climate actions.

This section examines a central assumption of existing theories of international climate cooperation. Standard models imply that countries' mitigation efforts should behave as strategic substitutes, so I examine whether this prediction is reflected in observed patterns of climate policymaking. I compare countries' adoption of climate laws and the stringency of their environmental policies to aggregate climate action abroad. These analyses should be interpreted as reduced-form conditional correlations of whether countries' actions behave as substitutes or complements, rather than as causal estimates of policy diffusion. Contrary to the prediction of strategic substitution, I find that countries' propensities to enact climate laws and pursue more stringent environmental policies are positively correlated. These results point toward a strategic complementarity in countries' climate actions rather than effort substitution.

²Strategic substitutability is conceptually distinct from decisionmakers' failure to internalize their negative externalities, or to "underprovide" effort, both of which are often invoked when discussing temptations to free-ride. However, these arguments require a normative optimum so that one can compare equilibrium effort levels to what is socially optimal, and thus demand significantly more theoretical and empirical structure.

To examine law adoption, I use data from the Climate Change Laws of the World project (Nachmany et al. 2017), which provides information on adopted climate laws from 1990 to 2024. The data covers laws adopted by the national governments of 199 countries plus the European Union. To be included in this dataset, a document must have full legal force or delineate a current set of government policy objectives, which pertain specifically to challenges of climate change.

For each country i , I observe the number of climate laws adopted in year t as well as the total count of laws adopted by all other nations besides i (notated as $-i$). To capture the correlation between country i 's climate law adoption and that of other nations, I run the following regression model via ordinary least squares with country-clustered standard errors:

$$\text{Laws}_{i,t} = \beta \log(\text{Other Laws}_{-i,t-1}) + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t}.$$

The dependent variable $\text{Laws}_{i,t}$ is either the logged count of laws that country i passed in year t , or a binary indicator for whether or not country i passed any laws in year t . The independent variable $\text{Other Laws}_{-i,t-1}$ is the count of laws adopted by all other nations around the globe besides i , which I lag by one year to avoid simultaneity concerns. Country fixed effects α_i capture any time-invariant factors that affect country i 's likelihood of climate policy adoption. I also include country-specific time trends $\lambda_{i,t}$ which should parse out any secular increases over time in countries' climate policymaking behavior or increases in demand for climate policy over time within countries. Additionally, these terms intend to separate out general equilibrium effects, or any global benefits to increased mitigation that could correlate with a contemporaneous rise in climate law adoption across countries.

In addition to lawmaking, I also examine policy stringency, a continuous and cumulative measure of efforts into climate remediation that probes the intensive margin of countries' policies. I use two measures of policy stringency, the OECD's Environmental Policy Strin-

gency (EPS) index (Botta and Koźluk 2014; Kruse et al. 2022) and the Climate Action Policies and Measurement Framework (CAPMF) developed by Nachtigall et al. (2024). The EPS covers 40 countries between 1990 and 2020 and ranges from 0 to 6, where greater values imply greater stringency. Stringency is defined as the ability to explicitly or implicitly place a price on pollution through market-based (taxes, trading schemes, feed-in tariffs, and deposit and refund schemes) and non-market policies (command-and-control standards and subsidies). The CAPMF runs between 1990 and 2023 and measures the breadth of mitigation policy actions for 49 countries and the European Union on a scale from 0 to 10.³

For country i in year t , I run the following regression via ordinary least squares with country-clustered standard errors:

$$\text{Stringency}_{i,t} = \beta \text{ Average Other Stringency}_{-i,t-1} + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

which fits country i 's policy stringency in year t as a function of the average policy stringency of all other nations besides i in year $t - 1$. I also include country fixed effects α_i and country-specific time trends $\lambda_{i,t}$.

The estimand of interest is the sign of β , which in reduced form captures the relationship between country i 's climate policymaking and the behavior of other countries. Extant theoretical approaches, in which mitigation efforts are substitutable, would expect $\beta < 0$: the passing of laws or more stringent policies by others reduces country i 's tendencies to pass its own laws or adopt its own stringent policies. In all specifications, I divide independent and dependent variables by their within-country standard deviations so that β targets the within-country correlation between country i 's behavior and the behavior of other nations.

Contra prior theoretical expectations, there is a *positive* and statistically significant cor-

³I focus on the CAPMF's measurement of the stringency of "sectoral policies" rather than "cross-sectoral policies" or "international frameworks" although results are robust to the inclusion of these policy variables as well.

	Laws (Count) (1)	Laws (Binary) (2)	EPS (3)	CAPMF (4)
log(Other Laws)	0.819*** (0.049)	0.867*** (0.053)		
Avg. Other Stringency			0.706*** (0.143)	0.818*** (0.051)
Observations	7,000	7,000	1,200	1,650
R ²	0.331	0.344	0.904	0.962
Within R ²	0.307	0.305	0.852	0.960
DV Mean	0.643	0.713	2.04	1.25
Number of Countries	200	200	40	50
Country fixed effects	✓	✓	✓	✓
Country $\times$ Year trends	✓	✓	✓	✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table 1: Effects of Previous Law Adoption and Policy Stringency on Climate Policymaking

relation across climate actions, as shown in Table 1. Across the board, increased effort by other nations, measured either as the adoption of laws or more stringent environmental policy, is associated with greater mitigation measures. A one standard deviation increase in the laws adopted by or the policy stringency of other nations is associated with between 0.7 and 0.9 standard deviations of one’s own adoption of climate actions, a substantively meaningful effect.⁴ Since this correlation is positive, it suggests a complementarity rather than a substitutability in the adoption of climate laws across countries. Results are robust to alternative time trend specifications (including year random effects), the exclusion of country fixed effects (targeting a cross-country correlation), the inclusion of time-varying controls, as well as weighting by GDP per capita or emission per capita; see Appendix B.

These results are suggestive of a complementary relationship between efforts across coun-

⁴The sample of nations included in the stringency dataset—mostly larger, more developed countries with a larger impact on the climate—are likely the cases where we may have expected to observe greater effort substitution, thereby reinforcing the importance of the result.

tries. We may still be worried about an omitted variable or some other general equilibrium effect that produces a positive correlation on average, even if the partial equilibrium relationship across countries is negative. In the specifications above, the country-year time trends attempt to correct for this concern; Appendix B.3 also examines an alternative empirical analysis exploiting U.S. presidential elections as discontinuities in expected global climate leadership. Consistent with the descriptive evidence presented here, countries do not respond to the election of a more climate-ambitious U.S. administration by decreasing their own efforts, nor do they respond to a less climate-ambitious presidency with increased effort, as an account of effort substitution would predict.

Taken together, these findings imply that free-riding cannot be the sole driver of cross-national mitigation, since other countries' efforts are positively associated with a nation's own climate action. Several mechanisms could generate such complementarities: foreign mitigation may lower the costs of subsequent action through economies of scale; intensify competition for green investment and industrial advantage; or generate political pressure for other countries to follow. I focus instead on an informational mechanism in which governments are uncertain about the economic returns to decarbonization and use foreign policy experience to learn about them. With this mechanism, I augment standard collective action models of international climate politics to show that learning can generate complementarities across countries; when its effects outweigh free-riding, produces positive correlations in climate policy.

Model

Consider sequential climate policymaking by $n = 2$ countries indexed by i . Each country chooses a level of climate mitigation effort $a_i \in \mathbb{R}$. Positive levels of effort $a_i > 0$ represent mitigation actions like renewable energy investments, while negative effort $a_i < 0$ can be

thought of as anti-climate measures like deforestation. As the game features sequential moves, country 1 (“he”) moves before country 2 (“she”), and country 2 observes country 1’s choice prior to taking an action.

Similar to conventional theories, countries’ abatement efforts generate global public goods but impose local costs. Let $A = a_1 + \lambda a_2$ be the weighted average of global efforts where $\lambda \geq 0$ parameterizes the relative weight of country 2’s subsequent efforts vis-à-vis country 1’s effort. This parameter can be interpreted, for example, as the relative size and scale between the two countries (i.e., the importance of burden-sharing) or as a measure of temporal discounting. Global effort generates the public good of emissions reductions valued at $g(A)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing function with diminishing marginal sensitivity away from zero. But adopting climate-friendly policies is costly—there are technological, financial, and political consequences from imposing costs on citizens or large domestic polluters to implement reforms—and exerting effort comes at a cost $c_i(a_i)$ where $c_i : \mathbb{R} \rightarrow \mathbb{R}_+$ is a convex function increasing in the magnitude of a_i . To keep the problem tractable,⁵ let

$$g(A) = \begin{cases} \sqrt{A} & \text{if } A \geq 0 \\ -\sqrt{|A|} & \text{if } A < 0, \end{cases}$$

and

$$c_i(a_i) = c_i|a_i|.$$

The payoff from mitigation efforts also depends on a fundamental uncertainty: the economic returns from decarbonization. Substantively, this captures the value of renewable energy industries, the creation of clean jobs, or the value of integration into green supply chains, which is unknown at time of investment. I model this uncertainty as an unknown

⁵These functional forms allow substantively for the inclusion of negative effort, rather than truncating effort at zero, but are also useful technically in allowing for feasible integration over negative-valued signals, obtaining a one-to-one mapping between signals and effort.

state of the world, θ . Neither country knows the true realization of θ , as countries do not know whether or not the green transition will ultimately pay off, but share the common prior that θ is normally distributed with mean μ and precision γ , $\theta \sim N(\mu, \frac{1}{\gamma})$. The mean μ captures all shared expectations about the benefits of green policy, which may include information from green markets, international reports, or other public sources. In addition, each country observes a noisy signal of the state of the world x_i that is normally distributed and centered around the true value of θ with precision β , $x_i \sim N(\theta, \frac{1}{\beta})$. Countries' signals are their private information and are thus their private beliefs about the economic returns to green policy. Since higher signals are on average more likely to indicate larger θ , a country with a higher signal is more "optimistic" about the economic benefits of decarbonization.⁶ Heuristically, these signals capture the idea that countries would only want to exert effort into mitigation if they are sufficiently optimistic that the investments would reap economic benefits.

Putting things together, country i 's utility is written as

$$u_i(a_i, A; \theta) = \theta g(A) - c_i(a_i).$$

Countries' utility functions preserve the free-rider logic of canonical collective action models: mitigation generates a global public good whose marginal value declines as total effort increases. Unlike existing models, however, countries are uncertain about the economic returns to green policies. Because country 2 observes country 1's action before choosing her own effort, actions may reveal information about these returns in addition to affecting the provision of the public good. Climate policies therefore influence subsequent decisions through two channels: a free-riding channel and a learning channel. The theoretical analysis

⁶Conversely, countries with strong negative signals would be more pessimistic about the economic benefits from mitigation and thus less willing to take action. This, along with the fact that countries' efforts a_i can be negative, captures policy retrenchment or rollback, permitting unfavorable information about the returns to decarbonization to generate downward policy adjustment within the same framework.

characterizes which of these effects dominates—formally studying the sign of $\frac{da_2}{da_1}$.

I examine weak Perfect Bayesian equilibria. Country 1’s strategy is a function $\alpha_1 : \mathbb{R} \rightarrow \mathbb{R}$ that maps his signal x_1 into an effort level a_1 . Country 2’s strategy is a function $\alpha_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ that maps her signal x_2 and country 1’s effort a_1 into an effort level a_2 . Beliefs about θ are formed via Bayes’s rule.

Comments on the Model

In blending collective action and observational learning models, I make a few simplifications that require further interpretation.

In contrast to extant observational learning models which allow the number of agents n to grow arbitrarily large (for a review, see [Bikhchandani et al. 2024](#)), I fix the number of countries at $n = 2$. I make this simplification because agents in these models are typically backward-looking as their behavior is only linked through the information conveyed by prior movers, essentially boiling down to a decision-theoretic problem. However, since countries’ actions exhibit forward-looking strategic interdependence in this model, increasing the number of countries adds mathematical complexity without developing further theoretical nuance. In Appendix F, I present a simplified model with more than two countries that is closer to the traditional observational learning setup. By way of interpretation though, we can think of the first mover as a bloc of countries in reduced form, while the second mover is an additional country considering how its effort affects global abatement on the margin.

In a departure from extant collective action models of international climate policymaking (e.g., [Kennard and Schnakenberg 2023](#)), I introduce two modifications: uncertainty about the economic benefits accrued from climate abatement efforts, and sequential moves. Both of these elements are necessary to model the learning mechanism. While these models often assume complete information, uncertainty about the economic benefits from green policy introduces the possibility to learn about a critical feature of the policy problem. In introducing

sequential moves, I treat the timing of countries' climate actions as exogenously determined. This is a simplifying assumption, and there are plausible foundations for endogenizing the order of moves. For instance, differences in the precision of countries' information, such as more reliable assessments of the economic or technological feasibility of a green transition, could lead better-informed countries to act earlier (e.g., [Zhang 1997](#)).

The model assumes that countries are uncertain about a common state governing the economic returns to decarbonization θ . Some common component is necessary for cross-national learning: if the benefits of green investment were entirely country-specific, foreign policy choices would contain no information about the returns to acting at home. The common state therefore captures the part of the green transition's economic value that governments can learn about from others' experiences.

One final interpretive comment bears on how learning occurs and the interpretation of countries' signals about the economic benefits from decarbonization x_i . The model abstracts from the institutional channels through which governments observe foreign policy experiences. In practice, diplomats, bureaucrats, and technical experts interact repeatedly through venues such as UNFCCC negotiations, where they exchange information about policy design, implementation, and outcomes ([Rietig 2025](#)). More generally, bureaucratic officials can carry information acquired through international interactions back into domestic policymaking ([Clark and Zucker 2024](#); [Beleli 2022](#)).

Theoretical Analysis

I begin by establishing the equilibrium of the game (Proposition 1). Then I demonstrate how learning effects can dominate free-riding concerns, yielding a positive correlation across countries' climate actions (Proposition 2).

Equilibrium Climate Efforts

To solve for the equilibrium of the model, proceed by backward induction and begin with country 2's problem. By equilibrium conjecture, country 1's strategy $a_1 = \alpha_1(x_1)$ exists and is one-to-one, meaning that for any level of effort a_1 that country 2 observes, she can infer the signal that generated it, $x_1 = \alpha_1^{-1}(a_1)$. She therefore combines this inferred signal with her own private information x_2 to form a posterior belief about the economic returns to decarbonization. Conditional on her signal x_2 and country 1's effort level a_1 , country 2 believes that $\theta|x_2, a_1 \sim N(\frac{\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2}{\gamma + 2\beta}, \frac{1}{\gamma + 2\beta})$. Hence, country 2's expected utility is

$$u_2(a_2; x_2, a_1) = E[\theta|x_2, a_1]g(a_1 + \lambda a_2) - c_2(a_2),$$

where her optimal mitigation effort $\alpha_2(x_2, a_1)$ solves the following first-order condition:

$$\underbrace{E[\theta|x_2, a_1]}_{\text{expected green benefits}} \cdot \underbrace{g'(a_1 + \lambda\alpha_2(x_2, a_1))\lambda}_{\text{marginal global contribution}} - \underbrace{c_2'(\alpha_2(x_2, a_1))}_{\text{marginal costs}} = 0.$$

Under the assumed functional forms, country 2's optimal effort is

$$\alpha_2(x_2, a_1) = \begin{cases} -\frac{1}{\lambda}a_1 + \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2} & \text{if } x_2 \geq -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1) \\ -\frac{1}{\lambda}a_1 - \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2} & \text{if } x_2 < -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1). \end{cases} \quad (1)$$

Country 2's optimal effort balances expected economic returns from green policy, the marginal contribution of her effort to global mitigation, and the domestic costs of climate action. Note that when $x_2 < -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1)$, country 2's posterior expectation of the returns to decarbonization is negative, $E[\theta|x_2, a_1] < 0$. In this case she exerts negative effort, corresponding to policies that impede rather than promote the green transition.

To preview some of the intuition for the main theoretical result, note that country 1's

effort a_1 affects country 2's optimal mitigation investment $\alpha_2(x_2, a_1)$ through two channels. First, country 1's effort has a negative direct effect of changing country 2's marginal contribution to global mitigation: if country 1 has already committed to large emissions reductions, then country 2's policies have a smaller marginal impact on global abatement, thereby creating a temptation to free-ride (the first term of $\alpha_2(x_2, a_1)$). In this way, country 1's behavior engenders a substitutability across nations' mitigation actions, which is reminiscent of classical collective action pathologies.

Moreover, country 1's effort also affects country 2 through a second, informational channel: her expectations about the benefits from green policies. When country 1 exerts more effort, it leads country 2 to believe that country 1's signal x_1 was higher, and thus more optimistic about the economic returns from decarbonization (given by $\alpha_1^{-1}(a_1)$ in the second term of $\alpha_2(x_2, a_1)$). Country 2 updates positively about the value of green policy, thereby learning from country 1, which is quantified by an increase in her posterior mean $E[\theta|x_2, a_1]$. This learning effect generates complementarity in the countries' climate remediation efforts.

Now consider the problem of country 1, who faces a similar decision but knows that his actions will influence the trajectory of global climate efforts. Given his signal x_1 , country 1 has a posterior belief about the state $\theta|x_1 \sim N(\frac{\gamma\mu+\beta x_1}{\gamma+\beta}, \frac{1}{\gamma+\beta})$ as well as country 2's signal $x_2|x_1 \sim N(\frac{\gamma\mu+\beta x_1}{\gamma+\beta}, \frac{2\beta+\gamma}{\beta(\beta+\gamma)})$. Since he does not know x_2 , country 1 does not know how much effort country 2 will commit to downstream, and thus has expected utility

$$u_1(a_1; x_1) = E_{x_2} \left[E[\theta|x_1] g(a_1 + \lambda \alpha_2(x_2, a_1)) \right] - c_1(a_1),$$

such that his optimal mitigation effort $\alpha_1(x_1)$ satisfies the following first-order condition:

$$E_{x_2} \left[\underbrace{E[\theta|x_1]}_{\text{expected green benefits}} \cdot \underbrace{g'(a_1 + \lambda \alpha_2(x_2, a_1))}_{\text{marginal global contribution}} \underbrace{\left(1 + \lambda \frac{d\alpha_2(x_2, a_1)}{da_1}\right)}_{\text{expected impact on country 2}} \right] - \underbrace{c'_1(a_1)}_{\text{marginal costs}} = 0.$$

Under the assumed functional forms, country 1's optimal effort is

$$\alpha_1(x_1) = \begin{cases} \frac{\lambda(\gamma\mu + \beta x_1)^2}{4c_1 c_2 (2\beta + \gamma)(\beta + \gamma)} & \text{if } x_1 \geq -\frac{\gamma\mu}{\beta} \\ \frac{-\lambda(\gamma\mu + \beta x_1)^2}{4c_1 c_2 (2\beta + \gamma)(\beta + \gamma)} & \text{if } x_1 < -\frac{\gamma\mu}{\beta}. \end{cases} \quad (2)$$

Proposition 1 summarizes the equilibrium. All formal proofs are relegated to Appendix A.

Proposition 1. *There exists a unique equilibrium characterized by the functions $\alpha_1(x_1)$, $\alpha_2(x_2, a_1)$ such that $a_1^* = \alpha_1(x_1)$, as defined in Equation 2, and $a_2^* = \alpha_2(x_2, a_1^*)$, as defined in Equation 1. Country 1 has beliefs $\theta|x_1 \sim N(\frac{\gamma\mu + \beta x_1}{\gamma + \beta}, \frac{1}{\gamma + \beta})$ and country 2 has beliefs $\theta|x_2, a_1^* \sim N(\frac{\gamma\mu + \beta\alpha_1^{-1}(a_1^*) + \beta x_2}{\gamma + 2\beta}, \frac{1}{\gamma + 2\beta})$.*

In equilibrium, country 1 recognizes that his mitigation effort affects country 2 through two channels. Greater effort reduces country 2's marginal contribution to global mitigation, creating incentives for her to free-ride. At the same time, because country 1 only undertakes costly climate investments when sufficiently optimistic about their economic returns, greater effort also signals more favorable private information about the prospects of an economically fruitful green transition. Country 1 must therefore weigh the possibility that greater effort induces downstream free-riding against its informational value in encouraging subsequent climate action.

This interaction creates strategic feedback: because country 1 knows that greater effort may induce downstream free-riding, a large mitigation investment becomes an especially strong signal that he believes the economic returns to decarbonization are high. The informational value of country 1's effort therefore partially offsets the strategic incentives to free-ride by the first mover that characterize conventional collective action models. Indeed, absent the informational value of mitigation, country 1 could maximally free-ride off of country 2's subsequent efforts, mirroring the collective action failures emphasized by conventional

models.⁷

Corollary 1 establishes the learning mechanism and summarizes the preceding discussion. Country 1's effort is informative to country 2 about the value of green policy: when country 1 is more optimistic, he exerts greater effort. Having observed greater effort, country 2's expectations about the economic returns to climate policy increase, which can motivate further effort downstream.

Corollary 1. *Country 1's climate policy is informative about θ :*

- *Country 1's equilibrium climate effort is increasing in its signal x_1 , $\frac{da_1^*}{dx_1} > 0$;*
- *Country 2's posterior expectation of θ is increasing in country 1's effort, $\frac{dE[\theta|x_2, a_1^*]}{da_1} > 0$.*

Complements and Substitutes, Learning and Free-Riding

I now characterize how free-riding and learning jointly determine countries' incentives to adopt climate policy. These two mechanisms operate simultaneously but push in opposite directions: free-riding encourages strategic substitution, whereas learning generates strategic complementarities. The key theoretical question is therefore whether country 2's equilibrium response to country 1's effort is positive or negative. Formally, I ask: what is the sign of $\frac{d\alpha_2(x_2, a_1)}{da_1}$?

Fix two possible effort levels by country 1, $a'_1 > a_1$, and recall that country 2's optimal effort level solves

$$\underbrace{E[\theta|x_2, a_1]}_{\text{expected green benefits}} \cdot \underbrace{g'(a_1 + \lambda\alpha_2(x_2, a_1))\lambda}_{\text{marginal global contribution}} = \underbrace{c'_2(\alpha_2(x_2, a_1))}_{\text{marginal costs}},$$

⁷In a complete information version of the model where θ is known to all players, country 1's equilibrium effort is $a_1^* = -\infty$, corresponding to maximal free-riding on country 2's subsequent mitigation effort.

such that country 2 trades off her beliefs about the economic benefits of decarbonization and her marginal contribution to global abatement efforts with the marginal domestic costs of investment.

The left-hand side of this equation has two components that are in tension with one another. Given country 1's efforts, country 2's marginal contribution to global abatement is smaller when country 1 has already invested large amounts of effort into providing the global benefit of mitigation, $g'(a'_1 + \lambda a_2) < g'(a_1 + \lambda a_2)$. There are smaller marginal gains from additional costly remediation, which tempts country 2 to free-ride off of country 1's efforts. This force engenders substitution across countries' climate policies because the effort exerted by country 1 discourages reforms from country 2.

However, country 1, having adopted costly climate reforms, signals belief in an economically valuable green transition to country 2, $E[\theta|x_2, a'_1] > E[\theta|x_2, a_1]$. Emboldening country 2 to take more ambitious climate action because she now holds more optimistic expectations about the value of reform, country 1's efforts generate complementarities in countries' behavior.

Corollary 2 formalizes this discussion by decomposing the effect of country 1's action on country 2's effort into these two constituent partial effects, where a_1 represents the direct returns from country 1's effort and $\alpha_1^{-1}(a_1)$ represents the informational value.

Corollary 2. *Country 2's optimal mitigation effort is:*

- *Decreasing in the direct effect of country 1's effort, $\frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} < 0$;*
- *Increasing in the informational effect of country 1's effort, $\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} > 0$.*

Figure 1 illustrates the effects of the free-riding and learning mechanisms, plotting country 2's effort $\alpha_2(a_1, x_2)$ as a function of country 1's effort a_1 given different possible signals x_2 . The first panel depicts the direct effect of increasing country 1's mitigation policies, holding

constant the informational value of such effort. As in traditional accounts, this begets less ambitious climate action from country 2 because her marginal contributions are diminished, generating free-riding temptations, indicated by the negative slope of the lines in the panel. By contrast, the second panel shows that, holding the direct effect constant, country 2 is incentivized to exert greater effort when country 1 does more. This is precisely because country 2 has more optimistic beliefs about the value of green policy upon observing that country 1 has exerted more effort; the positive slope of the lines in this panel reflects this learning.

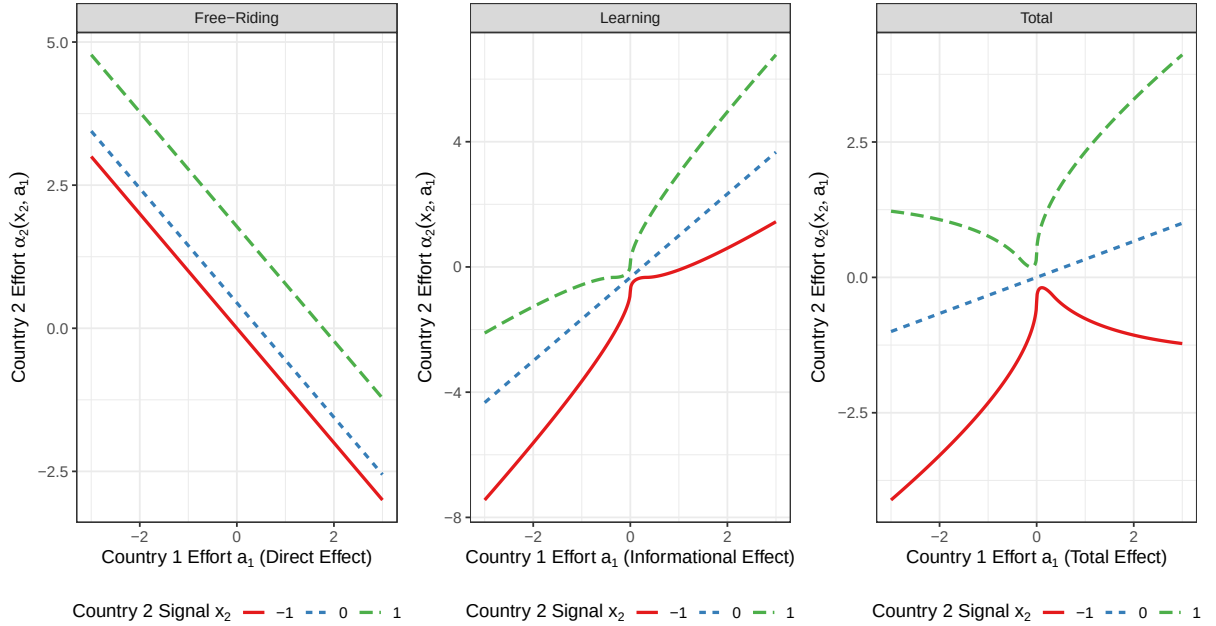


Figure 1: Substitutes and Complements Mechanisms on Climate Mitigation Efforts

$$\mu = 0.5, \gamma = 1, \beta = 1, \lambda = 1, c_1 = 0.5, c_2 = 0.25$$

The third panel of Figure 1 combines these two channels. In equilibrium, the direct and informational consequences of country 1's effort cannot vary independently: a marginal increase in a_1 simultaneously reduces country 2's marginal contribution to global mitigation and changes the information she infers from country 1's action. The slope of the lines in the panel therefore shows the total equilibrium effect, $\frac{d\alpha_2(x_2, a_1)}{da_1}$. When the learning effect

dominates, this effect is positive and countries' climate policies are strategic complements; when free-riding dominates, it is negative and their policies are strategic substitutes.

Proposition 2 determines the sign of the total effect. This tells us if countries' climate actions are strategic complements or substitutes in equilibrium, and thus if they are positively or negatively correlated.

Proposition 2. *Country 2's equilibrium mitigation effort is increasing in country 1's effort, $\frac{d\alpha_2(x_2, a_1)}{da_1} > 0$, when the following holds:*

$$\underbrace{\frac{|E[\theta|x_2, a_1]|}{|E[\theta|x_1]|}}_{\text{relative posterior beliefs}} > \underbrace{\frac{c_2}{\lambda c_1}}_{\text{relative costs}}. \quad (3)$$

Proposition 2 identifies the conditions under which the two mechanisms interact to generate complementary climate policymaking. The left-hand side compares the posterior assessments of the economic returns to decarbonization held by the two countries. Intuitively, complementarity is more likely when country 2 remains sufficiently convinced of the economic returns from mitigation policies after incorporating both her own information and what she learns from country 1's observed effort.

The right-hand side summarizes the relative costs and importance of the two countries' efforts. Greater implementation costs for country 2, c_2 , make complementarity less likely because favorable information produces a smaller downstream policy response. By contrast, greater costs for country 1, c_1 , make complementarity more likely because a given level of costly effort conveys stronger information about his optimism. A larger λ likewise makes complementarity more likely because country 2's subsequent effort contributes more heavily to global mitigation. The effects of prior and signal precision, γ and β , operate through the countries' posterior beliefs and need not be monotonic.

Under complementarity, the inequality demonstrates that the implications of the collective action problem for international climate cooperation are conditional. Free-riding incentives are always present: additional mitigation by other countries invariably reduces the marginal contribution of one’s own effort. But this need not dominate countries’ equilibrium behavior. When the informational value of foreign climate action is sufficiently large, countries’ mitigation efforts become mutually reinforcing rather than mutually substitutable. The central dilemma of international climate cooperation is therefore not the inevitability of collective action failures, but whether governments learn enough from one another’s experiences for the informational gains from foreign mitigation to outweigh free-riding temptations.

This result provides a theoretical interpretation of the positive cross-national correlations documented above. Such correlations can arise when the informational consequences of foreign climate action are strong enough that a marginal increase in one country’s effort induces greater rather than smaller mitigation abroad. When Equation 3 holds, learning outweighs the direct incentive to substitute away from foreign mitigation, producing complementary climate policymaking across countries.

Recent episodes of domestic political backlash to climate policies can also be understood in terms of their interaction with international policymaking incentives. Political backlash and electoral resistance to climate remediation—often spearheaded in developed democracies by right-wing parties (Stokes 2016; Colantone et al. 2024; Gazmararian 2025; Stutzmann 2025; Voeten 2025; Heddesheimer, Hilbig and Voeten 2026)—can raise the marginal costs of implementing mitigation policies c_2 . This increases the threshold that the strength of the learning effect must overcome in order for climate policies to be complementary across borders. Thus, a domestic political landscape that is more hostile to climate policy can shift incentives toward effort substitution: even if country 2 learns favorable information from country 1’s action, stronger domestic opposition may make that information insufficient to offset the incentive to free-ride.

To gain further empirical traction on the inequality described in Proposition 2, we would need to observe governments’ private signals—their beliefs about the returns to green policy—which are not directly measurable. I therefore proceed in two steps. I first document that governments increasingly treat foreign climate policies as sources of information about the economic consequences of decarbonization, thereby providing direct evidence of learning. I then evaluate several empirical implications of the learning mechanism using public opinion and cross-national policy data.

Documenting Climate Policy Learning

The model assumes that governments infer information about the economic returns to decarbonization by observing climate policies adopted abroad. Before evaluating the model’s broader empirical implications, it is therefore useful to establish that such learning occurs in practice. To measure this informational mechanism, I examine whether and how governments cite and engage with the policy experiences of other nations in their own climate plans. If governments are learning from one another—and, in particular, about the potential economic returns from pursuing climate policies—then one way this would manifest is as a justification for a country’s own climate policy adoption.

I draw on the full-text corpus of climate laws and policies compiled by Climate Policy Radar, which hosts the Climate Change Laws of the World Project ([Nachmany et al. 2017](#); [Institute 2025](#)), the same source underlying the law adoption data used above. The corpus provides the complete text of tens of thousands of documents enacted between 1990 and 2025, including national legislation, executive policies and action plans, international agreements, and climate litigation. I restrict attention to “laws and policies” concerning climate mitigation—documents authored by national governments that carry legal force or delineate current policy objectives. This yields a working corpus of 6,720 national laws and policy

documents spanning nearly two hundred countries and the European Union.

Within these documents, I use a keyword search to identify every instance in which a climate plan mentions another country, producing 108,782 candidate references to the policy experiences of other nations. The empirical task is to ascertain which of these references constitutes genuine engagement with the climate actions of other countries; I use a large language model classifier to determine whether such references reflect policy learning or are other, incidental mentions of other countries. Appendix C further describes the LLM, its coding scheme, and validation process.

I operationalize two potential measures of learning based on countries' climate policy citations of others. First is a broader, more encompassing measure that records whether a government substantively engages another country's mitigation policy or experience as an input to its own policymaking—describing its design or results, benchmarking against it, or explicitly drawing lessons from it. I identify 1,913 such learning citations, appearing in 629 documents; the citing governments span 138 countries and draw on the experiences of 117 others. Second, I obtain a more demanding measure that records whether a government specifically invokes the economic benefits wrought by the climate policies of another nation as a rationale for pursuing similar action at home—for example, a policy's consequences for jobs, lowered costs, investment, or market leadership in the enacting country. This measure maps most directly onto the paper's theoretical mechanism, in which observing green policy abroad raises a country's belief in the value of its own green transition; there are 790 citations of this kind across 337 documents.⁸ Additional summary statistics can be found in Appendix D. Table 2 provides examples illustrating how countries cite others' policy experiences, invoking the economic benefits that can come from green policies.

⁸As a check on whether these references primarily reflect learning rather than international economic competition, a prominent alternative explanation, I manually reviewed all 790 economic value citations. Only 2.5% explicitly framed climate policy adoption in terms of competition. This does not rule out competitive motives that are left unstated, but suggests that these citations are not predominantly expressions of international competition. See Table A.28 for examples.

Citing Country	Cited Country	Document (year)	Excerpt
Spain	Germany	Plan for Renewable Energy in Spain 2011-2020 (2011)	“The scheme ... [a] feed-in tariff ... is basically the same system in countries like Germany or Denmark that, together with Spain, outline the success scenarios for renewable energies ... [and aims] to guarantee a reasonable return on investments while the learning curves and economies of scale place the different technologies in optimal conditions to fight side by side with conventional sources.”
Germany	Austria	Recovery and Resilience Plan (2021)	“In Austria, the expansion of electromobility together with the provision of the necessary infrastructure has led to an increase in jobs ... We assume a similar effect in Germany, whereby the measure contributes to the dynamic labor market.”
Netherlands	United States	National Climate Agenda (2013)	“In the US ... it has been calculated that 1 US dollar in ... capacity that develops these [energy-efficiency] standards leads to 60,000 US dollars in additional relief of energy costs for companies and consumers. To accelerate this, the Netherlands advocates making additional capacity available ...”
Indonesia	Vietnam	Comprehensive Investment and Policy Plan (2023)	“Research conducted in Vietnam shows that replacing coal power with solar or wind will more than double the number of jobs on an average megawatt capacity.”
Colombia	Uruguay	Just Energy Transition Roadmap (2024)	“Uruguay, thanks to its Renewable Energy Law, is considered a global pioneer ... [It] currently supplies 92.8% of its electricity grid with renewable sources, serving as a perfect example worldwide of how the energy grid is ... shifting from importing energy to exporting surpluses...”

Table 2: Citations Invoking Economic Benefits of Another Country’s Climate Policy

This measurement strategy encodes an important selection concern. A citation records beliefs about the economic value of green policy only once a government has undertaken the effort of writing a climate plan; learning thus would be observed only among countries

who legislate themselves. Selection into law adoption is a limitation for interpreting raw levels of learning. That said, the countries most tempted to free-ride are precisely those that would not exert positive climate effort, so the citations we do observe likely understate how widely the experiences of others shape policy. Relatedly, there may be strategic selection into which countries cite and which get cited; however, the specificity of the measure to economic benefits should alleviate these concerns as the theory is agnostic as to why a particular country would be cited over another.

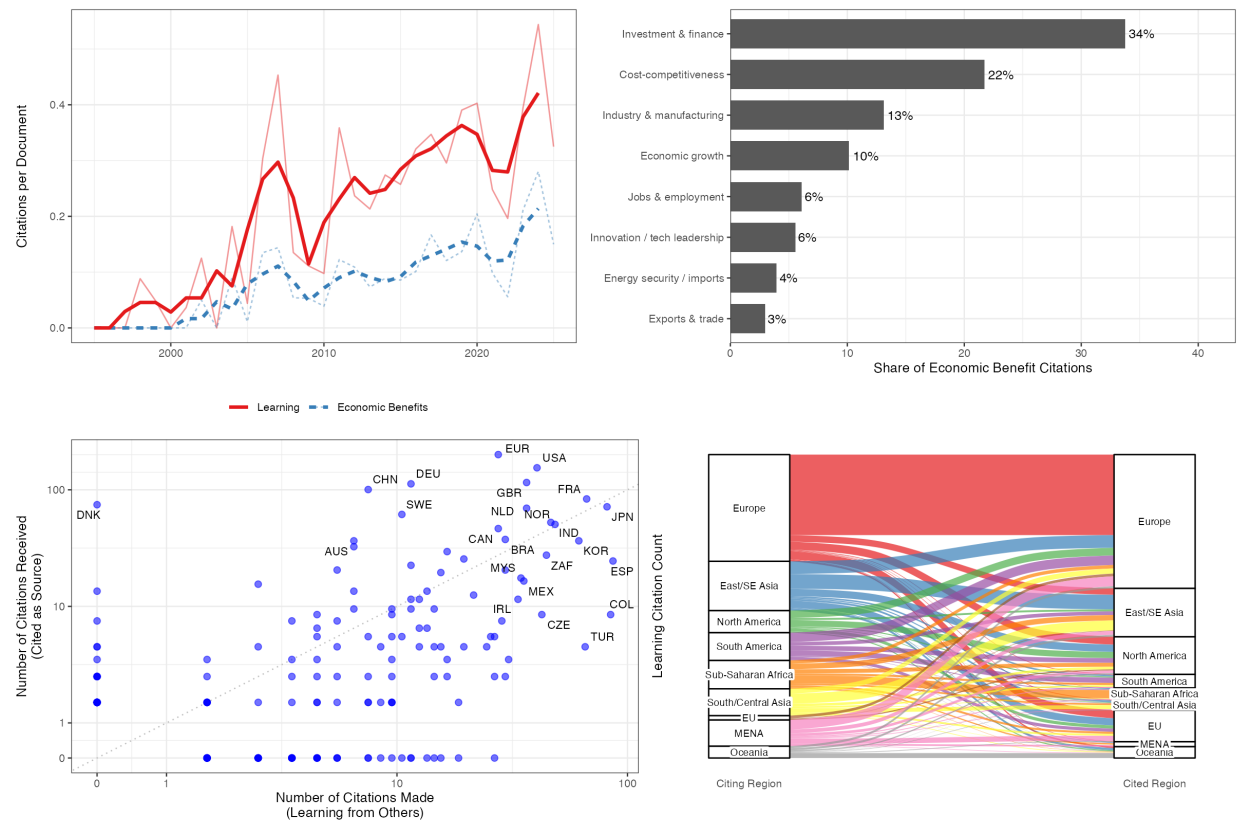


Figure 2: Citations of Other Countries in Climate Policy Documents

Figure 2 provides a broad overview of the trends in policy learning about the economic benefits of climate mitigation. Its four panels summarize, in turn, the growth of citations over time, the economic content governments invoke when they engage another country’s policy experiences, the distribution of countries as sources and citers of policy ideas, and the

geography of who learns from whom.

The upper-left panel traces the incidence of learning between 1995 and 2024, measured as the number of flagged citations per climate document (three-year moving average, raw trends are the lighter lines). The red solid line depicts the broader policy learning measure while the blue dashed line describes the citations that explicitly invoke the economic benefits of climate remediation. Over time, there has been a surge in policy citations, from an average of 0.024 citations per document in the mid-1990s to 0.34 in the early 2020s, thus an increase of over 1300%. Invocations of economic benefits when citing others' experiences have similarly exploded, with no mentions of these potential gains until the 2000s to approximately 0.21 citations per document in 2024. This trend holds even after normalizing by the global number of climate policies enacted each year, demonstrating that it is not an artifact of secular increases in climate policy adoption. Governments increasingly justify domestic mitigation by appealing to foreign policy experience, and they do so with growing emphasis on the economic consequences of decarbonization. Learning about the economic returns to climate policy has become an increasingly salient feature of international climate politics.

Central to the theoretical argument, governments invoke foreign climate policy based on their economic outcomes. Using a keyword search of the citations classified as those explicitly mentioning economic benefits, the upper-right panel sorts citations based on the content of the potential gains to be accrued. Governments mostly frame the economic value of the green transition in terms of investment and capital (34% of economic benefit citations) and cost competitiveness or cheaper energy prices (22% of citations). Appeals to industrial development and manufacturing (13% of citations) and to aggregate economic growth (10% of citations) follow. There are relatively fewer mentions of the creation of green jobs (6% of citations) and innovation and market leadership (6% of citations). Other cited benefits include energy security (4% of citations) and trade (3% of citations). The content of these citations shows that governments appear to engage with how mitigation can deliver domestic

economic gains, precisely the mechanism developed in the theoretical model.

Turning to the bottom-left panel, I examine the distribution of which countries are likely to cite others and which are likely to be cited. The horizontal axis depicts how often a country engages the experience of others, and the vertical axis charts how often its own experience is cited in return. Countries above the diagonal are net sources; this group is highly concentrated among a small group of states, demonstrating that learning is relatively asymmetric. Early movers and large economies—the European Union, Denmark, Germany, the United Kingdom, China, and, perhaps surprisingly, the United States—are disproportionately cited, while many smaller and later-transitioning economies primarily reference others. Countries below the diagonal are these net citers, drawing on others far more than they are drawn upon, with Colombia, Turkey, South Korea, and Spain as the clearest cases. A sizable group of 43 nations, responsible for roughly one in eight citations, cites the experience of others but is never cited in return. These pure learners are concentrated among smaller and later-transitioning economies—including Kazakhstan, Albania, Guatemala, and Panama. Their behavior is consistent with the notion of decarbonization as development: rather than shaping the frontier of climate policy, they primarily draw on foreign experience to assess the domestic economic opportunities associated with the green transition.

Finally, the lower-right panel studies the geography of learning to ask: who learns from whom? I aggregate dyadic citation patterns across countries into regional flows. The figure demonstrates that learning is strongly regionalized: the largest flows are internal to regions, with a substantial share of citations flowing to immediate neighbors. Overwhelmingly, the most climate policy learning occurs within Europe: 27% of all citations are between European countries, excluding any potential references to the European Union more broadly. Learning within East and Southeast Asia is the second most frequent occurrence, encompassing about 5% of total citations. Extraregional citations are concentrated on a few sources. Europe, North America, and the industrialized economies of East and Southeast Asia draw citations

from every region, and the European Union (separate from its member states) is cited far more often than it cites, acting as a near-pure source of policy ideas. Regions at earlier stages of the green transition, notably the Middle East and North Africa and South and Central Asia, appear primarily as citing regions rather than cited sources.

Taken together, these patterns suggest that governments increasingly treat foreign climate policies as evidence about the economic consequences of decarbonization. Cross-border references to other nation’s mitigation experiences serve as a useful measure to examine how nations justify the value of their own climate policies. As shown, learning is becoming more common over time, is often framed in terms of domestic economic gains, is concentrated on a relatively small number of policy leaders, and is strongest among geographically proximate countries. These findings do not establish that learning outweighs free-riding in equilibrium, but they provide direct evidence that the informational mechanism developed in the model is present in contemporary climate policymaking. The next section turns to the model’s empirical implications, asking whether public beliefs and subsequent policy adoption evolve in ways consistent with this mechanism.

Implications of the Learning Mechanism

The previous section provided direct evidence that countries are engaging in learning—reflected by citations of others’ policies in national climate documents—and increasingly pointing to the economic benefits of mitigation to justify climate policy. The model implies that these informational spillovers should shape both beliefs and subsequent policy choices. Corollary 1 implies a series of observable relationships that should arise whenever governments update on foreign climate policies. I evaluate three such implications, beginning with within-country relationships to increasingly demanding tests of cross-border learning.

1. More favorable assessments of the returns to green policy should translate into greater

effort within countries.

2. Observing more ambitious climate efforts abroad should lead to upward revisions in beliefs about the economic returns to mitigation across countries.
3. More favorable assessments of the returns to green policy abroad should induce greater climate action across countries.

Testing these implications thus provides a bridge between the formal model and observed patterns of international climate action, together tracing the logic of the learning mechanism. The first implication establishes that a country’s own climate efforts are informative of its beliefs. The second implies a cross-national updating effect. Finally, the third and most demanding test captures the reduced-form relationship of learning: optimistic signals foster greater effort at home, which in turn leads others to update their beliefs favorably and respond with stronger policies. Demonstrating this pattern would provide the strongest evidence of learning, though it is also the most difficult to isolate empirically, since strategic considerations such as free-riding may counteract downstream effort provision.

To probe these implications, I examine the relationship between beliefs and mitigation effort. Beliefs are measured using repeated cross-sectional Eurobarometer data to capture public opinion on the seriousness of climate change across 27 European countries.⁹ As shown above, the plurality of policy citations are European countries citing each other, making Europe a prime context for studying the presence of learning.¹⁰ Moreover, this data is dynamic, allowing for the study of changes in country’s beliefs and subsequent policies over time. I measure climate effort using the logged count of climate laws enacted in a particular year.

⁹While the cleanest theoretical interpretation of the signals in the model are beliefs held by elites, meta-studies have shown that gaps in mass and elite beliefs are likely overstated ([Kertzer 2022](#); [Kertzer and Renshon 2022](#)).

¹⁰While a substantively appropriate setting, examining learning within EU countries is also a more empirically difficult test of the mechanism given how Brussels maintains control over the trajectory of climate policy. Thus any effects captured in the design will represent marginal changes away from this baseline.

Between 2011 and 2023, Eurobarometer asks citizens to assess whether they believe that climate change is a serious problem on a scale from 1 to 10, where 1 is “not at all a serious problem” and 10 is “an extremely serious problem.”¹¹ Since climate effort is measured at the country-year level, I aggregate all respondents’ beliefs about the importance of climate change by taking the average within countries for each year. In all specifications, the dependent and independent variables are divided by their within-country standard deviation to more cleanly estimate correlations between climate actions and beliefs.

The first implication concerns the within-country relationship between beliefs and climate efforts, which I test by estimating

$$\text{Effort}_{i,t} = \beta \text{CC Serious}_{i,t-1} + \alpha_i + \lambda_t + \varepsilon_{i,t},$$

where α_i are country fixed effects, which account for baseline differences in assessments of climate change importance across countries. Since comparisons are entirely within countries, I estimate a set of year fixed effects λ_t , which account for secular changes in beliefs and policy effort over time—including EU-wide shifts in climate policy requirements. The theory predicts a positive relationship: if beliefs about the value of climate action shape policy choices, stronger assessments of seriousness should be associated with greater domestic mitigation effort.

The second implication asks whether climate policies implemented abroad lead governments and citizens to update their beliefs favorably. I estimate

$$\text{CC Serious}_{i,t} = \beta \text{Other Effort}_{-i,t-1} + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

where $\text{Other Effort}_{-i,t-1}$ is the sum of laws passed by all other EU member states besides

¹¹While this question does not perfectly map onto the model’s core uncertainty—whether green policies will be economically beneficial—it provides substantively meaningful indicators of how publics form expectations about climate policy demand and success.

i in the previous year. Country fixed effects and country-year time trends help to target the within-country effect of others' actions on beliefs. A positive coefficient would provide evidence that climate policies generate informational spillovers across countries consistent with complementarity.

The third implication, in which domestic beliefs correlate with downstream policy abroad, is tested via

$$\text{Effort}_{i,t} = \beta \text{Average Other CC Serious}_{-i,t-1} + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

and $\text{Average Other CC Serious}_{-i,t-1}$ is the average belief that climate change is a serious problem across all other EU countries besides i in the previous year. As before, country fixed effects and country-specific trends isolate within-country changes over time. A positive relationship would provide the strongest reduced-form evidence for the learning mechanism, linking foreign beliefs to increased downstream domestic climate effort.

The results of these three tests are found in Table 3. Each row of the table corresponds to the findings for a separate empirical implication of the learning mechanism. While the theory's implications become more stringent in ascending order, and evidentiary weight thus should be placed accordingly, we find suggestive evidence consistent with the learning mechanism across the board.

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	log(Own Laws)	0.157** (0.075)	263
2	log(Other Laws)	Own Seriousness	0.644*** (0.159)	390
3	Other Seriousness	log(Own Laws)	0.269*** (0.072)	265
p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$				
Robust standard errors clustered at the country level				

Table 3: Empirical Implications of the Learning Mechanism

The first implication establishes the domestic relationship between beliefs and climate

policymaking. Countries exhibiting greater public concern about climate change also adopt more climate laws: a one standard deviation increase in climate concern is associated with a 0.157 standard deviation increase in logged climate laws. This relationship is consistent with the model’s premise that more favorable beliefs about climate policy are associated with greater mitigation effort at home.

The second and third implications provide more diagnostic evidence on cross-border learning. The second row shows that greater climate effort abroad is associated with significantly higher domestic concern about climate change, consistent with citizens updating their beliefs after observing foreign policy experiences. The third row shows that more favorable beliefs abroad are themselves associated with stronger domestic climate policymaking. These effects are substantively meaningful: a one standard deviation increase in foreign climate effort is associated with a 0.644 standard deviation increase in domestic climate concern, while a one standard deviation increase in climate concern abroad corresponds to a 0.269 standard deviation increase in domestic climate law adoption.

In Appendix E, I include a number of ancillary specifications, like those that exclude time effects, include lagged dependent variables, and estimate relationships using a recoded 1-4 scale; results are broadly robust. I also use measures of policy stringency to operationalize effort, yielding mixed results. Most estimates are statistically indistinguishable from zero, although the CAPMF estimate for the third implication is negative and statistically significant.

The results are consistent with the theoretical expectations. Within countries, greater public concern predicts the adoption of more climate policies. Moreover, greater foreign climate effort is followed by more favorable domestic assessments, consistent with cross-border updating, and these beliefs are in turn associated with subsequent policy effort. These findings provide additional evidence for learning because belief updating is a direct implication of the learning mechanism, whereas free-riding or competition alone can generate correlated

actions without requiring systematic changes in beliefs. In canonical collective action models, foreign climate policies affect domestic effort by changing the strategic incentives to provide a global public good, and do not systematically alter beliefs. By contrast, the observed correlations between beliefs and actions, within and across countries, provide empirical support for the learning channel as a meaningful driver of international climate policy.

Conclusion

This paper develops an account of international climate policymaking in which foreign mitigation generates two consequences for other governments. By increasing the provision of a global public good, foreign climate action reduces the marginal value of additional effort and creates incentives to free-ride. At the same time, it reveals information about the economic returns to decarbonization, allowing governments to learn from policies adopted abroad. Since both free-riding and learning affect incentives for climate mitigation, the *ex ante* relationship for cross-national policy implementation is ambiguous. With theoretical analysis, I show how these channels jointly determine whether countries' climate efforts become strategic substitutes or complements. Empirically, I provide evidence that governments increasingly invoke foreign policy experiences when reasoning about the economic benefits of mitigation, and document that beliefs and policy efforts evolve across countries in ways consistent with the learning mechanism. The collective action problem therefore remains central to international climate politics, but its equilibrium implications are conditional: persistent incentives to free-ride need not produce substitutable policymaking when foreign climate action generates sufficiently strong informational spillovers.

While the analysis emphasizes learning as a mechanism generating complementarities in climate policymaking, there may be alternative explanations for policy diffusion. The paper does not argue or assume that learning is the only mechanism of diffusion at work ([Gra-](#)

ham, Shipan and Volden 2013; Cao et al. 2014). For example, foreign mitigation may lower technology costs through economies of scale (Gallagher 2014; Nemet 2019), governments may compete for green investment and industrial advantage (Meckling 2025), nations may seek international status (Keohane and Victor 2016) or emulate prominent policy leaders (Busby and Urpelainen 2020; Hale 2020), or even respond to coercive pressure from abroad (Bare and Kennard 2026). Technological change can alter the underlying returns to mitigation, while learning shapes governments’ beliefs about those returns. Competition over green industries, in turn, presupposes that governments recognize valuable rents worth capturing. Cross-border learning can therefore help explain how green economic opportunities become politically salient in the first place, after which competitive and reputational dynamics may reinforce policy diffusion. The evidence in this paper shows that governments explicitly draw lessons from foreign climate policies about their economic consequences and that foreign action is associated with changes in beliefs that predict subsequent mitigation effort. Distinguishing when learning gives way to, or interacts with, these mechanisms is an important task for future research on the diffusion of climate policies.

The argument also suggests a broader perspective on the role of international institutions in climate politics. Existing theories typically view climate agreements as mechanisms for mitigating free-riding through monitoring, enforcement, or reciprocity, treating collective action as the central obstacle to international cooperation (Hovi, Ward and Grundig 2015; Nordhaus 2015; McAllister and Schnakenberg 2022). The learning mechanism developed here demonstrates that when governments update their beliefs about the economic returns to decarbonization by observing one another’s climate policies, cooperation can emerge through decentralized policy diffusion rather than institutional coordination alone. This perspective helps explain why substantial uptake has occurred despite the comparatively weak institutional architecture of the global climate regime (Keohane and Victor 2011; Keohane and Oppenheimer 2016; Barrett 2021). International institutions may facilitate these informa-

tional processes by increasing transparency and encouraging the dissemination of policy experiences (Raiser et al. 2020; Kamil and Karlsson-Vinkhuyzen 2023; van Deursen and Gupta 2025), but they are not a necessary condition for learning to occur. Countries learn directly from one another’s observable actions, implying that international cooperation may depend as much on the diffusion of optimistic expectations as on the institutional management of free-riding.

The theoretical framework advanced in this paper might assist in studying public goods problems in other realms of international politics as well. The core tension of the model boils down to when countries should invest in global public goods as a function of their collective beliefs about the value of these investments and the temptations to free-ride off of others; when the former mechanism dominates the latter, then contributions are complementary. In collective security, military investments by one state may reduce allies’ incentives to contribute, but they may also reveal private assessments of the severity of a common threat. During emerging disease outbreaks, protective measures adopted abroad may create opportunities to free-ride while simultaneously conveying information about the severity of the disease and the value of costly prevention. In such settings, governments respond to foreign contributions not only because those actions change how much of a public good has already been provided, but also because they reveal information about whether further effort is worthwhile.

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Appendix: Learning and Free-Riding in International Climate Policymaking

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A Formal Proofs

Proof of Proposition 1. To solve for the equilibrium, I conjecture the existence of a strategy for country 1 $a_1 = \alpha_1(x_1)$ and assume that $\alpha_1(x_1)$ is one-to-one. Proceeding by backward induction first consider country 2's effort investment given her signal x_2 and country 1's effort a_1 . Given that $\alpha_1(x_1)$ is one-to-one, we have $x_1 = \alpha_1^{-1}(a_1)$ and so country 2's posterior update about θ is $\theta|x_2, a_1 \sim N(\frac{\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2}{\gamma + 2\beta}, \frac{1}{\gamma + 2\beta})$. Country 2 therefore solves

$$\max_{a_2} E[\theta|x_2, a_1]g(a_1 + \lambda a_2) - c_2(a_2).$$

Differentiating with respect to a_2 , country 2's first-order condition is

$$E[\theta|x_2, a_1]g'(a_1 + \lambda a_2)\lambda - c_2'(a_2) = 0.$$

Note that if $E[\theta|x_2, a_1] < 0$, which occurs when $x_2 < -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1)$ then $E[\theta|x_2, a_1] < 0$ and country 2 exerts effort in the opposite direction. Given the functional form specifications and solving the above first-order condition, we have the following piecewise function:

$$\alpha_2(x_2, a_1) = \begin{cases} -\frac{1}{\lambda}a_1 + \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2} & x_2 \geq -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1) \\ -\frac{1}{\lambda}a_1 - \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2} & x_2 < -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1) \end{cases}$$

It is clear from the second-order condition that for any x_2 and any a_1 , $\alpha_2(x_2, a_1)$ is unique as the second-order condition is always negative:

$$E[\theta|x_2, a_1]g''(a_1 + \lambda a_2)\lambda^2 - c_2''(a_2) < 0.$$

Now consider country 1's effort choice. Given his own signal x_1 , he believes that $\theta|x_1 \sim N(\frac{\gamma\mu + \beta x_1}{\gamma + \beta}, \frac{1}{\gamma + \beta})$ and that country 2's signal $x_2|x_1 \sim N(\frac{\gamma\mu + \beta x_1}{\gamma + \beta}, \frac{2\beta + \gamma}{\beta(\gamma + \beta)})$. Let $m = \frac{\gamma\mu + \beta x_1}{\gamma + \beta}$ and $z = \sqrt{\frac{\beta(\gamma + \gamma)}{2\beta + \gamma}}$. Further, denote $q = \frac{\lambda(\gamma\mu + \beta\alpha_1^{-1}(a_1) + \beta x_2)^2}{4c_2^2(2\beta + \gamma)^2}$ and $t = -\frac{\gamma\mu}{\beta} - \alpha_1^{-1}(a_1)$. By backward induction, country 1's problem is to maximize

$$\max_{a_1} \int_{-\infty}^t [mg(-q)z\phi(z(x_2 - m))] dx_2 + \int_t^{\infty} [mg(q)z\phi(z(x_2 - m))] dx_2 - c_1(a_1).$$

Differentiating with respect to a_1 , country 1's first-order condition is

$$\begin{aligned} FOC &= mg(0)z\phi(z(t - m))\frac{dt}{da_1} + \int_{-\infty}^t -mg'(-q)\frac{dq}{da_1}z\phi(z(x_2 - m)) dx_2 \\ &\quad - mg(0)z\phi(z(t - m))\frac{dt}{da_1} + \int_t^{\infty} mg'(q)\frac{dq}{da_1}z\phi(z(x_2 - m)) dx_2 - c_1 = 0 \\ &= \int_{-\infty}^t \frac{m\beta\lambda}{2c_2(2\beta + \gamma)} \frac{1}{\alpha_1'(\alpha_1^{-1}(a_1))} z\phi(z(x_2 - m)) dx_2 \end{aligned}$$

$$\begin{aligned}
& + \int_t^\infty \frac{m\beta\lambda}{2c_2(2\beta + \gamma)} \frac{1}{\alpha_1'(\alpha_1^{-1}(a_1))} z\phi(z(x_2 - m)) dx_2 - c_1 = 0 \\
\Leftrightarrow \quad \alpha_1'(\alpha_1^{-1}(a_1)) &= \frac{(\gamma\mu + \beta x_1)\beta\lambda}{2c_2c_1(2\beta + \gamma)(\gamma + \beta)}.
\end{aligned}$$

It is clear that since q is increasing in $\alpha_1^{-1}(a_1)$, country 1's optimal strategy does not contain any "flat spots" as it is always optimal for him to induce greater effort from country 2. Observe that for $x_1 < -\frac{\mu\gamma}{\beta}$, $m < 0$ and so country 1 would then exert effort in the negative direction. By equilibrium conjecture, $a_1 = \alpha_1(x_1)$ so $\alpha_1^{-1}(a_1) = x_1$ and integrating with respect to x_1 yields

$$\alpha_1(x_1) = \frac{\beta\lambda x_1(2\gamma\mu + \beta x_1)}{4c_1c_2(2\beta^2 + 3\beta\gamma + \gamma^2)} + C.$$

The constant of integration is pinned down by the boundary condition that, at $x_1 = -\frac{\gamma\mu}{\beta}$, we have $E[\theta|x_1] = 0$. The equilibrium effort is thus

$$\alpha_1(x_1) = \begin{cases} \frac{\lambda(\gamma\mu + \beta x_1)^2}{4c_1c_2(2\beta + \gamma)(\beta + \gamma)} & x_1 \geq -\frac{\gamma\mu}{\beta} \\ \frac{-\lambda(\gamma\mu + \beta x_1)^2}{4c_1c_2(2\beta + \gamma)(\beta + \gamma)} & x_1 < -\frac{\gamma\mu}{\beta}. \end{cases}$$

Note that this is one-to-one in x_1 , confirming that $\alpha_1(x_1)$ is one-to-one in equilibrium. This means that $\alpha_1^{*-1}(\cdot)$ is well-defined so country 2 knows $x_1 = \alpha_1^{*-1}(a_1)$ in equilibrium.

Finally, observe that the second order condition is

$$-\frac{m\beta\lambda}{2c_2(2\beta + \gamma)} \frac{\alpha_1''(\alpha_1^{-1}(a_1))}{(\alpha_1'(\alpha_1^{-1}(a_1)))^3} < 0,$$

so the solution $\alpha_1(x_1)$ is the unique maximizer of country 1's utility. □

Proof of Corollary 1. Immediate given the equilibrium strategy of country 1:

$$\begin{aligned}
\frac{d\alpha_1(x_1)}{dx_1} &= \frac{\beta\lambda|\gamma\mu + \beta x_1|}{2c_1c_2(2\beta^2 + 3\beta\gamma + \gamma^2)} \geq 0. \\
\frac{dE[\theta|x_2, a_1]}{da_1} &= \frac{\beta}{2\beta + \gamma} \frac{1}{\frac{d\alpha_1(x_1)}{dx_1}} \geq 0.
\end{aligned}$$

Note that the absolute value in $\frac{d\alpha_1(x_1)}{dx_1}$ is simply because if $\gamma\mu + \beta x_1 < 0$, then we are on the lower branch of $\alpha_1(x_1)$ and the negative signs cancel. □

Proof of Corollary 2. Given country 2's equilibrium effort function as defined in Equation 1,

$$\frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} = -\frac{1}{\lambda} < 0.$$

$$\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} = \frac{\lambda \beta |\gamma \mu + \beta \alpha_1^{-1}(a_1) + \beta x_2|}{2c_2^2(2\beta + \gamma)^2} > 0.$$

Note that the absolute value in $\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)}$ comes from the piecewise nature of $\alpha_2(x_2, a_1)$, recall that if $x_2 < -\frac{\gamma \mu}{\beta} - \alpha_1^{-1}(a_1)$ then $E[\theta|x_2, a_1] < 0$. $\square$

Proof of Proposition 2. The total derivative of $\alpha_2(x_2, a_1)$ with respect to a_1 is

$$\frac{d\alpha_2(x_2, a_1)}{da_1} = \frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} + \frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} \cdot \frac{d\alpha_1^{-1}(a_1)}{da_1}.$$

From Corollary 2, we know that $\frac{\partial \alpha_2(x_2, a_1)}{\partial a_1} = -\frac{1}{\lambda} < 0$ for any x_2 . Similarly, $\frac{\partial \alpha_2(x_2, a_1)}{\partial \alpha_1^{-1}(a_1)} = \frac{\lambda \beta |\gamma \mu + \beta \alpha_1^{-1}(a_1) + \beta x_2|}{2c_2^2(2\beta + \gamma)^2} > 0$. By the inverse function theorem, $\frac{d\alpha_1^{-1}(a_1)}{da_1} = \frac{1}{\alpha_1'(\alpha_1^{-1}(a_1))}$, and from the derivation of country 1's optimal effort, we know $\alpha_1'(\alpha_1^{-1}(a_1)) = \frac{\beta \lambda |\gamma \mu + \beta \alpha_1^{-1}(a_1)|}{2c_2 c_1(2\beta + \gamma)(\gamma + \beta)}$. Imposing $\alpha_1^{-1}(a_1) = x_1$ and simplifying yields

$$\frac{d\alpha_2(x_2, a_1)}{da_1} = -\frac{1}{\lambda} + \frac{|\gamma \mu + \beta x_1 + \beta x_2|}{c_2(2\beta + \gamma)} \cdot \frac{c_1(\beta + \gamma)}{|\gamma \mu + \beta x_1|}.$$

Hence $\frac{d\alpha_2(x_2, a_1)}{da_1} > 0$ iff

$$\frac{|\gamma \mu + \beta x_1 + \beta x_2|}{|\gamma \mu + \beta x_1|} > \frac{c_2(2\beta + \gamma)}{\lambda c_1(\beta + \gamma)},$$

or

$$\frac{|E[\theta|x_2, a_1]|}{|E[\theta|x_1]|} > \frac{c_2}{\lambda c_1}.$$

$\square$

B Cross-Country Climate Actions: Robustness and Ancillary Analysis

Statistic	N	Mean	St. Dev.	Min	Max
Climate Laws (Count)	7,200	0.643	1.003	0.000	4.937
Climate Laws (Binary)	7,200	0.713	1.014	0.000	4.305
Other Laws	7,200	191.593	206.168	5	832
EPS	1,240	2.043	1.215	0.000	4.998
Avg. Other EPS	1,612	1.780	0.809	0.590	2.942
CAPMF	1,700	1.252	1.009	0.000	3.860
Avg. Other CAPMF	1,762	2.184	1.688	0.104	4.956

Table A.1: Summary Statistics for Climate Laws and Climate Policy Stringency

The main results employ country $\times$ year time trends. Table A.2 shows robustness of the findings without any temporal controls.

	Laws (Count)		Laws (Binary)		EPS		CAPMF	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(Other Laws)	0.530*** (0.012)	0.819*** (0.049)	0.526*** (0.011)	0.867*** (0.053)				
Avg. Other Stringency					0.918*** (0.013)	0.706*** (0.143)	0.972*** (0.003)	0.818*** (0.051)
Observations	7,000	7,000	7,000	7,000	1,200	1,200	1,650	1,650
R ²	0.302	0.331	0.314	0.344	0.900	0.904	0.960	0.962
Within R ²	0.277	0.307	0.273	0.305	0.845	0.852	0.958	0.960
DV Mean	0.643	0.643	0.713	0.713	2.04	2.04	1.25	1.25
Number of Countries	200	200	200	200	40	40	50	50
Country fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Country $\times$ Year trends		✓		✓		✓		✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.2: Effects of Previous Law Adoption and Policy Stringency on Climate Policymaking (with and without Time Trends)

The main results include a country fixed effect to parse out any time-invariant factors that lead countries to adopt laws or pursue environmental policies in a heterogeneous fashion. However, in so doing, the regression coefficient β thus targets the within-country correlation of the effect of other nations' climate actions on the adoption of climate laws. Table A.3 removes country fixed effects and shows that the results still hold.

	Laws (Count)		Laws (Binary)		EPS		CAPMF	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
log(Other Laws)	0.530*** (0.012)	0.803*** (0.048)	0.525*** (0.012)	0.849*** (0.052)				
Avg. Other Stringency					0.904*** (0.013)	0.409** (0.152)	0.971*** (0.003)	0.801*** (0.051)
Observations	7,000	7,000	7,000	7,000	1,200	1,200	1,650	1,650
R ²	0.267	0.270	0.257	0.262	0.529	0.536	0.909	0.911
Adjusted R ²	0.267	0.270	0.257	0.262	0.528	0.536	0.909	0.910
DV Mean	0.643	0.643	0.713	0.713	2.04	2.04	1.25	1.25
Year trends		✓		✓		✓		✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.3: Effects of Previous Law and Policy Stringency on Climate Policymaking (without Country Fixed Effects)

B.1 Climate Laws

In the main text, I employ country $\times$ year time trends to capture the effects of potential secular increases in the demand for climate policy. Table A.4 shows that results are robust to different types of time trends. Specifically, I estimate linear, quadratic, and cubic yearly time trends. I also estimate year random effects.

	Laws (Count)			Laws (Binary)		
	(1)	(2)	(3)	(4)	(5)	(6)
log(Other Laws)	0.819*** (0.048)	0.968*** (0.047)	0.388*** (0.064)	0.867*** (0.052)	0.911*** (0.052)	0.315*** (0.064)
Observations	7,000	7,000	7,000	7,000	7,000	7,000
R ²	0.305	0.310	0.330	0.319	0.319	0.340
Within R ²	0.280	0.285	0.306	0.278	0.279	0.301
Country fixed effects	✓	✓	✓	✓	✓	✓
Time trends	Linear	Quadratic	Cubic	Linear	Quadratic	Cubic

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.4: Effects of Climate Laws with Alternate Time Trends

	Laws (Count)	Laws (Binary)
	(1)	(2)
(Intercept)	−1.019*** (0.213)	−0.963*** (0.192)
log(Other Laws)	0.530*** (0.050)	0.526*** (0.040)
Country fixed effects	✓	✓
AIC	17652.524	17938.139
BIC	19043.818	19329.433
Log Likelihood	−8623.262	−8766.069
Num. obs.	7000	7000
Num. groups: year	35	35
Var: year (Intercept)	0.082	0.052
Var: Residual	0.655	0.685

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A.5: Effects of Climate Laws with Year Random Effects

While the results between other nations' laws and the tendency to adopt laws should be treated as descriptive or correlational, I also estimate these models with the inclusion of some time-varying controls. These control variables are meant to parse time-varying variation away from nations' climate policymaking behavior. Specifically, I estimate the revised model

$$\text{Laws}_{i,t} = \beta \log(\text{Other Laws}_{-i,t-1}) + X'_{i,t-1}\gamma + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

where the term $X'_{i,t-1}\gamma$ captures these controls. I include a lagged dependent variable, GDP per capita, population, and a country's growth rate (all from the World Bank), and the size of a country's winning coalition to proxy for regime type ([Bueno de Mesquita and Smith 2022](#)). Results are shown in Table A.6.

	Laws (Count)		Laws (Binary)	
	(1)	(2)	(3)	(4)
log(Other Laws)	0.557*** (0.033)	0.005 (0.076)	0.544*** (0.035)	0.224*** (0.079)
Lagged DV	0.071*** (0.019)	0.023 (0.019)	0.041** (0.016)	0.003 (0.017)
log(GDP per capita)	0.148 (0.101)	-0.097 (0.196)	0.168 (0.104)	0.027 (0.227)
log(Population)	0.089 (0.092)	-0.246 (0.655)	0.141 (0.105)	0.154 (0.552)
Growth	0.004 (0.003)	0.002 (0.003)	0.005* (0.003)	0.003 (0.004)
Winning Coalition Size	-0.419** (0.174)	-0.766*** (0.249)	-0.242 (0.183)	-0.447* (0.244)
Observations	5,075	5,075	5,075	5,075
R ²	0.366	0.396	0.366	0.389
Within R ²	0.345	0.376	0.331	0.355
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Controls	✓	✓	✓	✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.6: Effects of Climate Laws with Controls

We may be worried that only richer countries or only countries that are emissions-intensive need to pass laws. Tables A.7 and A.8 weight results by countries' GDP per capita and emissions per capita.

	Laws (Count)		Laws (Binary)	
	(1)	(2)	(3)	(4)
log(Other Laws)	0.613*** (0.011)	0.004 (0.076)	0.597*** (0.011)	0.248*** (0.077)
Observations	5,863	5,863	5,863	5,863
R ²	0.350	0.381	0.355	0.379
Within R ²	0.327	0.360	0.318	0.343
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GDP per capita	GDP per capita	GDP per capita	GDP per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.7: Effects of Climate Laws (Weighted by GDP per capita)

	Laws (Count)		Laws (Binary)	
	(1)	(2)	(3)	(4)
log(Other Laws)	0.506*** (0.031)	0.677*** (0.100)	0.498*** (0.031)	0.752*** (0.097)
Observations	6,545	6,545	6,545	6,545
R ²	0.293	0.329	0.301	0.335
Within R ²	0.257	0.295	0.245	0.282
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GHG per capita	GHG per capita	GHG per capita	GHG per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.8: Effects of Climate Laws (Weighted by Emissions per capita)

B.2 Policy Stringency

In the main text, I employ country $\times$ year time trends to capture the effects of potential secular increases in the demand for climate policy. Table A.9 shows that results are robust to different types of time trends. Specifically, I estimate linear, quadratic, and cubic yearly time trends. I also estimate year random effects.

	EPS			CAPMF		
	(1)	(2)	(3)	(4)	(5)	(6)
Average Other Stringency	0.705*** (0.141)	0.698*** (0.137)	0.339 (0.224)	0.818*** (0.051)	0.823*** (0.055)	0.646*** (0.047)
Observations	1,200	1,200	1,200	1,650	1,650	1,650
R ²	0.901	0.901	0.902	0.962	0.962	0.963
Within R ²	0.847	0.847	0.848	0.960	0.960	0.961
Country fixed effects	✓	✓	✓	✓	✓	✓
Time trends	Linear	Quadratic	Cubic	Linear	Quadratic	Cubic

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.9: Effects of Policy Stringency with Alternate Time Trends

	EPS	CAPMF
	(1)	(2)
(Intercept)	−0.546*** (0.078)	−0.327*** (0.041)
Average Other Stringency	0.916*** (0.015)	0.966*** (0.017)
Country fixed effects	✓	✓
AIC	1320.345	−540.609
BIC	1539.218	−253.957
Log Likelihood	−617.172	323.305
Num. obs.	1200	1650
Num. groups: year	30	33
Var: year (Intercept)	0.003	0.008
Var: Residual	0.148	0.033

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table A.10: Effects of Policy Stringency with Year Random Effects

While the results between other nations' stringency and the country i 's policy stringency should be treated as descriptive or correlational, I also estimate these models with the inclusion of some time-varying controls. These control variables are meant to parse time-varying variation away from nations' climate policymaking behavior. Specifically, I estimate the revised model

$$\text{Stringency}_{i,t} = \beta \text{Average Stringency}_{-i,t-1} + X'_{i,t-1}\gamma + \alpha_i + \lambda_{i,t} + \varepsilon_{i,t},$$

where the term $X'_{i,t-1}\gamma$ captures these controls. I include a lagged dependent variable, GDP per capita, population, and a country's growth rate (all from the World Bank), and the size of a country's winning coalition to proxy for regime type ([Bueno de Mesquita and Smith 2022](#)). Results are shown in Table A.11.

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Average Other Stringency	0.184*** (0.031)	0.134** (0.056)	0.213*** (0.034)	0.080*** (0.028)
Lagged DV	0.745*** (0.024)	0.738*** (0.024)	0.767*** (0.034)	0.753*** (0.036)
log(GDP per capita)	0.158*** (0.053)	0.529* (0.262)	0.107*** (0.025)	-0.097 (0.090)
log(Population)	0.238* (0.125)	-0.389 (0.924)	0.143*** (0.037)	-0.873* (0.438)
Growth	-0.006* (0.003)	-0.005 (0.003)	0.002 (0.001)	-0.0002 (0.001)
Winning Coalition Size	0.057 (0.242)	0.116 (0.510)	0.045 (0.091)	-0.234 (0.145)
Observations	1,165	1,165	1,469	1,469
R ²	0.952	0.954	0.977	0.978
Within R ²	0.927	0.930	0.975	0.977
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Controls	✓	✓	✓	✓

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.11: Effects of Policy Stringency with Controls

We may be worried that only richer countries or only countries that are emissions-intensive have greater policy stringency. Tables A.12 and A.13 weight results by countries' GDP per capita and emissions per capita.

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Average Other Stringency	0.923*** (0.013)	0.732*** (0.137)	0.968*** (0.005)	0.796*** (0.052)
Observations	1,170	1,170	1,523	1,523
R ²	0.898	0.902	0.958	0.961
Within R ²	0.845	0.852	0.956	0.959
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GDP per capita	GDP per capita	GDP per capita	GDP per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.12: Effects of Policy Stringency (Weighted by GDP per capita)

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Average Other Stringency	0.927*** (0.011)	0.785*** (0.131)	0.972*** (0.004)	0.782*** (0.052)
Observations	1,200	1,200	1,617	1,617
R ²	0.903	0.907	0.958	0.960
Within R ²	0.854	0.859	0.956	0.959
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓
Weights	GHG per capita	GHG per capita	GHG per capita	GHG per capita

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.13: Effects of Policy Stringency (Weighted by Emissions per capita)

Table A.14 shows robustness to measuring the behavior of other nations at the median rather than the average policy stringency, which eases concerns about extreme values.

	EPS		CAPMF	
	(1)	(2)	(3)	(4)
Median Stringency	0.919*** (0.013)	0.746*** (0.168)	0.972*** (0.003)	0.877*** (0.051)
Observations	1,200	1,200	1,650	1,650
R ²	0.900	0.904	0.962	0.963
Within R ²	0.844	0.850	0.960	0.961
Country fixed effects	✓	✓	✓	✓
Country × Year trends		✓		✓

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Robust standard errors clustered at the country level

Table A.14: Effects of Median Policy Stringency

B.3 U.S. Presidential Elections and Global Climate Action

As an additional test, I examine the effects of United States presidential elections on the adoption of climate laws. As the leader of one of the globe’s largest emitters, the identity of the American president heavily skews the trajectory of climate policy worldwide: in 2016, Donald Trump campaigned on withdrawing the United States from the Paris Agreement and retrenching the U.S. from the global climate regime, while Joe Biden’s 2020 campaign promoted American engagement in climate policymaking. The outcome of the 2016 (2020) election thus represents a negative (positive) perturbation to other countries’ best responses and subsequently informs their optimal decisionmaking in light of expected American behavior following the election. The classical logic predicts that other nations should increase (decrease) their adoption of climate laws following Trump’s (Biden’s) victory.

To capture the effects of the elections, I run a regression discontinuity in time at the country-month level using the following local linear regression specification:

$$\text{Laws}_{i,m} = \beta \mathbb{1}(m > t_k) + \varphi f(m - t_k) + \alpha_i + \varepsilon_{i,m}.$$

The dependent variable is either, as in the first empirical test, a binary indicator for whether country i adopted a climate law in month m or the logged count of climate laws adopted by country i in month m . As the running variable is time, the independent variable of interest is an indicator of whether time at month m has exceeded t_k , the calendar date of the election (either November 8 2016 or November 3 2020), and $f(\cdot)$ is a smooth function of time, which is measured in days until the election. Fixed effects at the country level α_i are also included to capture country-specific idiosyncrasies in law adoption. I report bias-corrected estimates with optimal bandwidth selection, weight observations using a triangular kernel (Calonico, Cattaneo and Titiunik 2014), and cluster standard errors by country.

The sign of β identifies the sign of the local average treatment effect of the U.S. election on other nations’ adoption of climate laws right at the date of the election.¹ Accordingly, this estimate should be thought of as countries’ expectations of the value of future effort given the winner of the election. A free-riding argument would expect the sign of β to be positive (negative) in 2016 (2020). In this account, Trump’s apathy toward climate policies increases the need for other nations to compensate, while Biden’s willingness to act decreases the incentives for others to exert costly effort.

Figure A.1 presents graphical evidence of a change in countries’ propensities to adopt climate laws around the 2016 and 2020 United States elections.² I fit a linear polynomial as well as a loess smoother on each side of the election cutoff. The left panel of the figure suggests that there is a small reduction in the adoption of climate laws when Donald Trump wins the election in 2016. By contrast, the right panel documents a large positive surge in

¹Given the design, bias in the LATE would emerge if countries strategically adopt climate laws before or after the election. For example, in 2016, the relevant bias in favor of the free-riding argument is if countries were to delay adoption of climate laws until after the election, attenuating the LATE. However, such a case seems unlikely as adoption would only become more difficult with a U.S. president who is less favorable to climate action in power.

²RD plots use the logged count outcome; plots for the binary dependent variable display similar results.

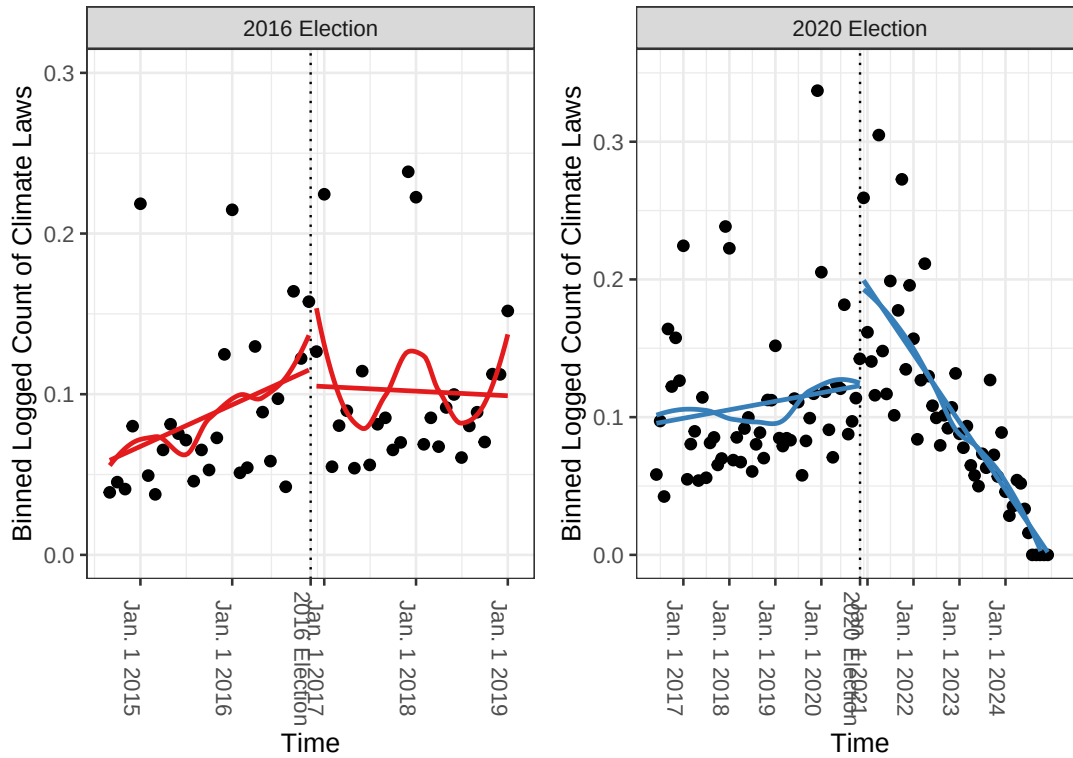


Figure A.1: RD Plots of U.S. Presidential Elections on Climate Law Adoption

the number of laws adopted when Joe Biden wins the election in 2020.

Table A.15 formalizes these estimates, documenting a negative, but statistically indistinguishable from zero, effect in 2016, and a positive and statistically significant effect in 2020. In both cases, the sign of the RD estimate refutes a story of strategic substitutes. That is, the data do not show that countries found it in their interest to do more when the U.S. signaled it would do less by electing Trump, nor would other countries do less when the U.S. elected Biden. When an anti-climate leader was elected in 2016, other countries followed suit and enacted fewer climate laws. In 2020, with the emergence of a pro-climate U.S. presidential winner, the international community responded with more climate action.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.009 (0.017)	-0.020 (0.019)	0.067*** (0.022)	0.070*** (0.020)
DV	Count	Binary	Count	Binary
DV Mean	0.094	0.109	0.105	0.116
Bandwidth	809.875	781.846	1616.020	1504.237
Effective Observations	10547	10149	20497	19701

Table A.15: RD Estimates of U.S. Presidential Elections on Climate Law Adoption

Results are robust to the inclusion of month fixed effects—which account for the influence of factors like the signing of new international environmental treaties, the onset of a global pandemic, or secular increases in worsening climate that may demand more action—as well as alternative specifications of the running variable, alternative bandwidths, placebo cutoffs, and sorting.

Table A.16 estimates the RD effect of U.S. elections without country fixed effects.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.009 (0.017)	-0.020 (0.019)	0.067*** (0.022)	0.070*** (0.020)
DV	Count	Binary	Count	Binary
Bandwidth (days)	810.142	782.037	1616.285	1504.298
Effective Observations	10547	10149	20497	19701

Table A.16: RD Estimates without Country Fixed Effects

The RD specification in the main text does not adjust for any other temporal shocks. Table A.17 controls for month fixed effects. These terms capture factors like signing of international agreements, the onset of global pandemic, or other time shocks that correlated with the adoption of climate laws around the election.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.014 (0.017)	-0.025 (0.019)	0.065*** (0.022)	0.066*** (0.020)
DV	Count	Binary	Count	Binary
Country fixed effects	✓	✓	✓	✓
Month fixed effects	✓	✓	✓	✓
Bandwidth (days)	829.619	824.242	1610.132	1510.144
Effective Observations	10746	10746	20298	19701

Table A.17: RD Estimates with Month Fixed Effects

Table A.18 shows that the main results hold if we define the running variable as months until the election instead of days.

	Trump (2016)		Biden (2020)	
	(1)	(2)	(3)	(4)
RD Election Effect	-0.005 (0.016)	-0.015 (0.018)	0.073*** (0.024)	0.078*** (0.022)
DV	Count	Binary	Count	Binary
DV Mean	0.091	0.105	0.117	0.129
Bandwidth	30.714	31.677	51.471	47.863
Effective Observations	12139	12537	17711	16915

Table A.18: RD Estimates with Monthly Running Variable

Figures A.2 and A.3 serve as placebo tests for the RD effects that vary the cutoff in the running variable. I look at the 90 day period around each election. The estimate highlighted in red is the actual estimate.

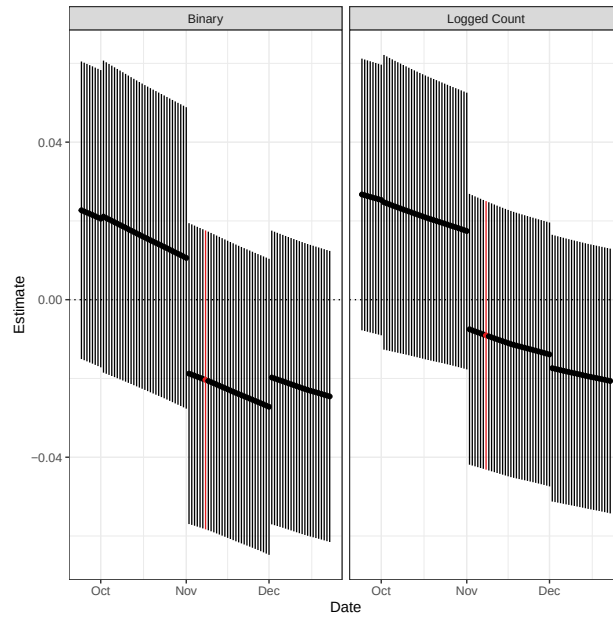


Figure A.2: RD Placebo Test (Trump)

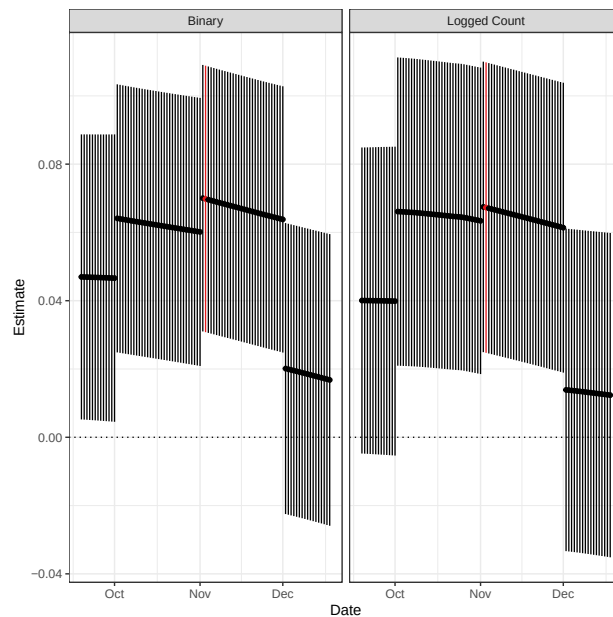


Figure A.3: RD Placebo Test (Biden)

Table A.19 shows the results of two tests of the no-sorting assumption, one based on McCrary (2008) and another based on Cattaneo et al. (2018). In both tests, the null hypothesis is that of no sorting. While the McCrary test is marginally significant for the Trump discontinuity, these results are suggestive of a lack of sorting in climate laws before/after the two U.S. elections.

Test	Election	p -value
McCrary (2008)	2016	0.091
McCrary (2008)	2020	0.149
Cattaneo et al. (2018)	2016	0.985
Cattaneo et al. (2018)	2020	0.931

Table A.19: RD Sorting Checks

Figure A.4 also shows how RD estimates are robust to different bandwidths. The estimate highlighted in red is the actual estimate. The Biden estimates are clearly robust; some of the Trump estimates change sign with very small bandwidths but they are not statistically significant.

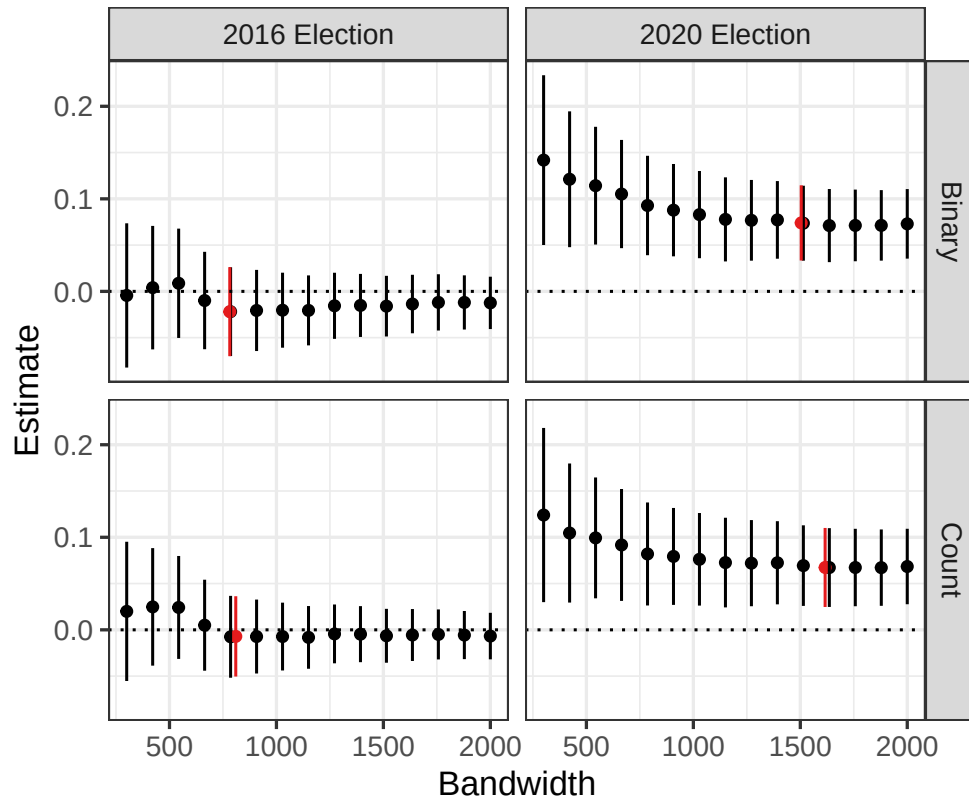


Figure A.4: RD Estimates with Alternative Bandwidths

C LLM-Based Classification of Learning Citations

This appendix describes how I construct and validate the two text-based measures of learning. The procedure has three stages—candidate generation, classification by a large language model (LLM), and validation against hand-coding—each detailed below. The full classifier prompt is reproduced at the end of the appendix.

Beginning from the full-text corpus of the Climate Change Laws of the World project and Climate Policy Radar (Nachmany et al. 2017; Institute 2025), restricted to national laws and policies in English,³ I scan every text block for mentions of another country using an exhaustive list of country names and case-sensitive acronyms with word-boundary matching. I discard self-references, boilerplate transposition of European Union and European Economic Area law, and substring artifacts. This yields 108,782 candidate references—passages in one country’s climate law that name another country—each retained together with its surrounding textual context.

I classify each candidate with a large language model. The primary classifier is Anthropic’s Claude Sonnet (model `claude-sonnet-5`), accessed through the Message Batches API on July 21 2026, with extended reasoning disabled and the default sampling temperature. Following common practice, I instruct the model to act as “an expert coder of climate-policy documents,” present it with the flagged passage and its surrounding context, and require it to return a single structured JSON object recording the three labels below.

Each candidate is coded on three nested indicators. The first, a *policy reference*, records whether the passage genuinely refers to the cited country’s own mitigation or clean energy policy, program, target, or outcome; it excludes mentions in which the country figures only as a trade or energy partner, a funder or donor, a bibliographic string, a currency unit, a geopolitical actor, or one entry in a descriptive roster. The second, a *learning citation*—the text’s first measure of learning—records whether the citing government substantively engages that policy or experience as an input to its own policymaking—describing its design or results, benchmarking against it, or drawing an explicit lesson from it. The third, an *economic value citation*, records whether a learning citation’s economic content—jobs, investment, cost, market or technological leadership, or a finance instrument—is stated in the passage itself or can be inferred. The domain is restricted throughout to climate mitigation and the clean energy transition; adaptation and other domains are excluded.

C.1 Validation

I follow standard practice for validating an LLM annotator against expert human coding (Gilardi, Alizadeh and Kubli 2023; Törnberg 2024). The codebook/prompt was developed iteratively on two initial samples and then frozen. Reliability is estimated on a fresh, held-out sample of 320 candidates (an initial 120 plus a further 200). Because instances of learning are relatively rare, the sample is stratified to contain enough positive cases (it holds 81

³Non-English documents are machine-translated by the data provider; I analyze the English-language text.

policy references, 38 learning citations, and 20 economic value citations), and all estimates are reweighted to corpus prevalence. I measure agreement with Cohen’s κ , together with precision and recall, and attach stratified-bootstrap 95% confidence intervals.

To guard against idiosyncrasies of any single model, I re-code the entire validation sample with a second, independently developed model family, OpenAI’s GPT-5 (model `gpt-5`), under the identical prompt and coding scheme. Agreement between the two families is reported in Table A.21.

Table A.20 reports the results for both model families.⁴ Agreement is moderate on every tier and, importantly, consistent across the two independently developed models. The economic value tier—the measure closest to the theory—is high-precision and moderate-recall for the classifier used at scale (precision 0.77, recall 0.48): it reliably flags explicit economic value citations but misses cases in which the economic content sits only in the surrounding context. The economic value counts reported in the main text should therefore be read as conservative lower bounds. Raw, non-chance-corrected agreement between the classifier and the coder is considerably higher—85%, 92%, and 96% on the three tiers—but κ is the appropriate metric given the rarity of the positive classes, since it removes the agreement expected by chance.

Tier	Claude (<code>claude-sonnet-5</code>)			GPT-5
	κ [95% CI]	Precision	Recall	κ [95% CI]
Policy reference	0.62 [0.51, 0.71]	0.70	0.74	0.58 [0.48, 0.68]
Learning	0.58 [0.42, 0.72]	0.69	0.58	0.57 [0.42, 0.69]
Economic value	0.57 [0.34, 0.75]	0.77	0.48	0.36 [0.15, 0.54]

Table A.20: Agreement with expert coding on the initial sample

Two further checks support the measure. First, the two model families agree with each other at least as closely as either agrees with the human coder—on 88%, 93%, and 96% of labels across the three tiers (Table A.21)—indicating that the coding scheme is not an artifact of a single model.⁵ Second, the classifier’s agreement with the human coding is insensitive to implementation choices: varying whether extended reasoning is enabled and whether requests are batched changes κ only at the margin (Table A.22). Where the classifier and coder disagreed, disagreements were re-adjudicated against the frozen codebook with full context. Applied to the full corpus, the classifier labels 1,913 learning citations and 790 economic value citations, the counts analyzed in the main text.

⁴Table shows Cohen’s κ , precision, and recall against a single expert coder on a held-out, stratified sample of 320 candidates (81 policy reference, 38 learning, 20 economic value positives), reweighted to corpus prevalence; 95% intervals from a stratified bootstrap. Both families reach moderate agreement on every tier; Claude is stronger on economic value citations.

⁵Table A.22 shows Cohen’s κ against the expert coding on the 120-candidate development sample; the adopted configuration’s agreement on the full $n = 320$ sample is reported in Table A.20. The choice of configuration barely affects agreement.

Tier	Raw agreement	Cohen’s κ [95% CI]
Policy reference	88%	0.70 [0.61, 0.78]
Learning	93%	0.64 [0.48, 0.76]
Economic value	96%	0.61 [0.38, 0.79]

Table A.21: Cross-model agreement (Claude vs. GPT-5)

Configuration	Policy reference	Learning	Economic value
Sonnet, extended reasoning on	0.54	0.48	0.55
Sonnet, extended reasoning off	0.58	0.43	0.68
Sonnet, batched, reasoning off (adopted)	0.56	0.46	0.73

Table A.22: Robustness to classifier configuration

Three limitations bear on interpretation. First, reliability is moderate ($\kappa \approx 0.57$ – 0.62 across tiers); economic value citations, though the most theory-relevant class, are the hardest to code, so moderate agreement is expected rather than a defect. Second, because recall for economic value citations is roughly 0.48, the measure is a conservative lower bound, and counts of levels of learning are best read as “at least.” Third, validation rests on a single expert coder—mitigated by the cross-model check and full-context adjudication.

C.2 Classifier Prompt

The model received the following instructions, codebook, and boundary examples, appended to each candidate passage (with the passage’s citing country, flagged country, document title and year, sentence, and surrounding context filling the bracketed fields).

Two nested tiers. Tier 1 (`learning_citation`): does country i engage country j ’s climate/energy policy or experience as something to *learn from*? Tier 2 (`economic_value_citation`, the paper’s θ): does that learning citation bear on the *economic value or feasibility* of j ’s green transition—whether stated explicitly or implied by an economic mechanism the passage invokes? Infer semantically; that is the reason to use a model here, not keyword matching. Nesting: `economic_value_citation` $\Rightarrow$ `learning_citation` $\Rightarrow$ `references_country_policy`.

Instructions. You are an expert coder of climate-policy documents. You are shown one passage from country {citing_country}’s national climate/energy law or policy. A regex flagged that it mentions {cited_country}. Judge only the reference to {cited_country}. Read the whole *Context*, not just the *Sentence*: the country name is often in one sentence (a membership list, a passing mention) while the learning or economic signal sits in an adjacent one. Decide from the full context, never from the flagged sentence alone. Return only a JSON object with these fields:

- `references_country_policy` (bool): a genuine reference to {cited_country}’s

own climate/energy policy, program, target, experience, or outcome? False if the country appears only as a trade or energy partner, transit point, war or sanctions actor, funder or donor, bibliographic string, currency unit, name collision, or one entry in a descriptive or enumerative list with no engagement of its policy. A list explicitly framed as learning sources (“learn from / examples such as A, B, C”) does count, for each named country.

- **country_actually_referenced** (ISO3 or “MULTIPLE”).
- **direction** (“i_learns_from_j” | “j_learns_from_i” | “peer/none”).
- **function**: **learning** · **compliance** (legal transposition/alignment with *j*’s or the EU rulebook) · **cooperation** (joint project, no lesson) · **trade_energy** · **geopolitics** · **descriptive_list** · **spurious**.
- **domain** (“mitigation” | “adaptation” | “both” | “other”). Only mitigation, clean energy, and the green transition are in scope; adaptation or other $\Rightarrow$ **learning_citation** = false.
- **learning_citation** (bool): true if **references_country_policy**, **direction** = **i_learns_from_j**, and the passage substantively engages the content of *j*’s policy, program, or its experience/effects as input to *i*’s own policy discussion. Explicit lesson-drawing language is sufficient but not required; describing *j*’s instrument, design, or results, or benchmarking against *j*, all count. False for out-of-scope domains; for compliance or accession obligations (“in line with the EU’s vision/acquis/directive” *alone*); for incidental data about *j* not tied to a policy; and for trade, energy supply, geopolitics, *j* as funder/donor, reverse direction, or a bare enumerative list.
- **economic_value_citation** (bool): true if **learning_citation** and the economic content is present in the passage itself. Counts: an explicit jobs, GDP, investment, or cost figure or claim; a stated “market/technology leader, champion, or pioneer”; a stated competitiveness or cost-saving gain; a named experience curve or documented deployment trajectory used as a feasibility model; deployment framed as investment; or a finance instrument whose economic function is the point. Does not count if the only economic angle is a generic chain the coder supplies (“efficiency $\Rightarrow$ saves money,” “any target $\Rightarrow$ growth”), emissions facts with no economic content, or bare “accelerate deployment.” Test: if you cannot quote the specific words carrying the economic claim, it is Tier 1, not Tier 2.
- **econ_reasoning** (string): if **economic_value_citation**, name the economic mechanism inferred, so the inference is auditable.
- **evidence_quote** (string): the ≤ 25 -word span justifying the labels.
- **confidence** (“high” | “med” | “low”).

D Documenting Learning: Ancillary Analysis

Table A.23 summarizes the corpus and the two learning measures. Of 108,782 candidate references across 6,720 national laws and policies, the classifier identifies 1,913 learning citations in 629 documents and, within them, 790 economic-value citations in 337 documents. The learning network spans 138 citing countries drawing on 117 distinct sources across 1,065 directed country pairs.

Quantity	Value
Candidate references (country mentions)	108,782
Total national laws and policies	6,720
Time span	1991–2025
Learning citations	1,913
in documents	629
Economic value citations	790
in documents	337
Citing countries	138
Cited sources	117
Directed country pairs	1,065

Table A.23: Summary statistics for the learning measure

Table A.24 ranks countries by the two roles that Figure 2 plots against one another: how often a country engages the experience of others, and how often its own experience is cited in return. The most active learners are middle-income and smaller advanced economies—Spain, Colombia, Japan, France, Turkey—while the most-cited sources are the early movers and large economies, led by the European Union, the United States, the United Kingdom, Germany, and China. Table A.25 lists the largest directed country pairs; ties to the European Union and among European neighbors dominate.

(a) Most active learners				(b) Most-cited sources		
Country	Cites	Countries Cited	Cites/Doc	Source	Cited	Citers
Spain	86	28	0.60	European Union	200	58
Colombia	84	32	1.14	United States	154	62
Japan	81	22	0.74	United Kingdom	115	51
France	66	15	0.41	Germany	112	45
Turkey	65	27	0.97	China	100	49
South Korea	61	17	0.67	France	83	37
India	48	18	0.50	Denmark	74	38
Norway	46	17	0.51	Japan	71	36
South Africa	44	30	0.75	Netherlands	69	39
Czechia	42	16	1.45	Sweden	61	28
United States	40	13	0.44	Norway	52	32
United Kingdom	36	16	0.23	India	50	33
Netherlands	36	14	0.44	Canada	46	23
Mexico	35	20	0.34	Brazil	37	27
Malaysia	34	15	1.48	Australia	36	25

Table A.24: Most active learners and most-cited sources (top 15)

Citing	Cited	Documents	with Economic value citations
Turkey	European Union	19	17
France	Netherlands	13	6
Japan	United States	13	9
South Korea	European Union	13	6
Colombia	European Union	11	6
Spain	France	11	6
United Kingdom	European Union	11	8
India	United States	10	5
Japan	European Union	10	9
Montenegro	European Union	9	4
Norway	Sweden	9	1
South Korea	Japan	9	5
Canada	United States	8	5
Croatia	Slovenia	8	5
France	Germany	8	4

Table A.25: Largest directed country pairs (top 15).

Table A.26 ranks countries by economic value citations alone. The set of countries invoking economic benefits in their citations is not identical to the set of learners overall: Japan and France, in particular, rise to the top, reflecting a greater tendency to frame others' policies in economic terms. On the receiving side, the sources are the large economies whose green transitions carry the clearest economic signal—the European Union, the United States, Germany, and China.

(a) Cite the economic value of others		(b) Cited for economic value	
Country	Economic value cites	Cites/Doc Source	Economic value cites
Japan	51	0.46 European Union	106
France	44	0.27 United States	72
Colombia	42	0.57 Germany	53
Spain	37	0.26 China	50
Türkiye	35	0.52 United Kingdom	47
Mexico	25	0.25 France	33
South Korea	24	0.26 Denmark	27
South Africa	24	0.41 Norway	27
India	21	0.22 Japan	24
Norway	20	0.22 Netherlands	24
United Kingdom	17	0.11 South Korea	20
Croatia	17	0.37 Brazil	18

Table A.26: Economic value citations, by country (top 12)

About a third of the corpus is machine-translated by the data provider, raising the concern that translation degrades the signal. Table A.27 splits the learning citations by whether the citing document was translated. Learning appears in both strata; the per-document rate is somewhat higher in English-original text (0.66 versus 0.46). This gap is consistent with some loss of signal in translation, which would make the measure conservative for non-English documents.

Table A.27: Learning citations by document language.

Documents	Learning	Economic value	Learning / Doc
English-original	1,233	501	0.66
Machine-translated	680	289	0.46

We may be worried that policy citations that invoke economic benefits of decarbonization could reflect international competition rather than policy learning. I manually reviewed the 790 economic benefit citations and found that only 20 of them, or 2.5%, explicitly mention international competition. Table A.28 provides examples from this small subset.

Citing Country	Cited Country	Document (year)	Excerpt
Norway	Singapore	Better Growth, Lower Emissions Strategy (2017)	“A number of countries are now developing policies to promote green competitiveness. In Singapore, the authorities are pursuing an active policy to develop, create markets for and deploy their own green technology. Canada is seeking to diversify its economy, and has set clear goals for its green growth policy to result in cuts in greenhouse gas emissions, more jobs and economic growth.”
India	United States	National Mission for Electric Mobility Plan 2020 (2012)	“In order to be globally competitive, it is essential for the Indian companies to reach capability levels comparable to US, Japan, China and Korea. Globally, Japan, United States and Korea have strong capabilities in lithium batteries. Japan leads with 5500 patents for battery cells and BMS and Korea and US follow suit with 2500 2000 and patents respectively.”
United States	China	Federal Sustainability Plan: Catalyzing America’s Clean Energy Industries and Jobs (2021)	“The European Union, the United Kingdom, and Canada have prepared their domestic industries and workers for the energy transition through industrial policy strategies. Furthermore, China’s domestic industry support has made it the global clean energy manufacturing leader. Moreover, the United States has become reliant on other countries for procuring clean energy technologies of the future, including zero-emission vehicles and certain renewable energy technologies.”

Table A.28: Citations Invoking Economic Benefits: International Competition

E Empirical Implications: Robustness and Ancillary Analysis

This appendix presents robustness and ancillary specifications for the mass beliefs analysis in the main text.

E.1 Mass Beliefs

Statistic	N	Mean	St. Dev.	Min	Max
Belief CC Serious	390	24.050	10.808	8.474	52.239
Own Laws	390	0.954	1.005	0.000	3.996
Other Laws	390	2.176	0.961	0.480	4.539

Table A.29: Summary Statistics for Mass Beliefs

Tables A.30-A.32 demonstrate that the results presented in the main text are robust to alternative specifications, namely the inclusion of a lagged dependent variable, the exclusion of time effects (either year fixed effects or country-year trends), and the use of a recoded 1-4 survey scale.

Implication	Regressor	Outcome	Estimate (SE)	<i>N</i>
1	Own Seriousness	log(Own Laws)	0.129 (0.096)	210
2	log(Other Laws)	Own Seriousness	0.536*** (0.122)	362
3	Other Seriousness	log(Own Laws)	0.294*** (0.090)	212

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.30: Empirical Implications of the Learning Mechanism (including Lagged DVs)

Implication	Regressor	Outcome	Estimate (SE)	<i>N</i>
1	Own Seriousness	log(Own Laws)	0.287*** (0.061)	263
2	log(Other Laws)	Own Seriousness	1.114*** (0.204)	390
3	Other Seriousness	log(Own Laws)	0.284*** (0.050)	265

p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.31: Empirical Implications of the Learning Mechanism (without Time Effects)

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	log(Own Laws)	0.151* (0.074)	263
2	log(Other Laws)	Own Seriousness	0.663*** (0.159)	390
3	Other Seriousness	log(Own Laws)	0.266*** (0.074)	265

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.32: Empirical Implications of the Learning Mechanism (Recoded 1-4 Scale)

The main text measures effort with the count of laws; Tables A.33 and A.34 use the CAPMF (Nachtigall et al. 2024) and EPS (Botta and Kořluk 2014; Kruse et al. 2022) measures of policy stringency instead. The results are weaker and generally fall short of conventional levels of statistical significance. However, there is a negative and significant relationship between other nations' seriousness and one's own policy stringency if stringency is measured using the CAPMF. As this is a substantively demanding test, it may be reflective of the other strategic forces like free-riding at play.

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	Own Stringency	−0.004 (0.020)	322
2	Avg. Other Stringency	Own Seriousness	−0.305 (0.199)	390
3	Other Seriousness	Own Stringency	−0.138*** (0.032)	324

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.33: Empirical Implications of the Learning Mechanism (CAPMF)

Implication	Regressor	Outcome	Estimate (SE)	N
1	Own Seriousness	Stringency	6.6×10^{-5} (0.167)	189
2	Avg. Other Stringency	Own Seriousness	−0.036 (0.066)	306
3	Other Seriousness	Stringency	0.160 (0.165)	189

p -values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Robust standard errors clustered at the country level

Table A.34: Empirical Implications of the Learning Mechanism (EPS)

F Alternative Model with $n > 2$ Countries

The model in the main text assumes a two-country interaction. The theoretical purchase of this approach was that it allowed for a parsimonious study of forward-looking incentives to exert effort or not. In this section, I describe a model that features $n > 2$ countries but remove this strategic interdependence. Similar to observational learning models (Banerjee 1992; Bikhchandani, Hirshleifer and Welch 1992; Smith and Sørensen 2000), countries are only backward-looking in their valuation of contributions to global effort.

F.1 Model Setup

Consider sequential climate policymaking between n countries indexed by $i = 1, \dots, n$ who decide whether to pursue climate reforms $a_i = 1$ or not $a_i = 0$. The action $a_i = 1$ represents climate reforms or policies instituted to facilitate a green transition in country i , while $a_i = 0$ represents maintaining the status quo. These countries take actions in a fixed order and can observe the choices of all countries before them.

Countries' payoffs to climate reform depend on two uncertain elements: the global benefit to a green transition and the private domestic cost of implementing reforms. The global benefit to a green transition is a binary state variable $\theta \in \{0, 1\}$. No country knows the true realization of θ —whether or not the green transition pays off or will be “successful” is unknown—but share the common prior $P(\theta = 1) = \pi \in (0, 1)$. If a country does not take climate action, it receives a payoff of zero. By taking climate action, country i receives a benefit normalized to 1 only if $\theta = 1$: this captures the idea that countries only want to pursue climate reforms if it is appropriate to do so or if the green transition is sufficiently likely to be successful.

While the benefits of a green transition are state dependent, their costs are not: choosing $a_i = 1$ comes at a cost $c_i \sim U[0, 1]$. These costs represent the domestic political feasibility of the green transition. Country i 's costs of implementation are privately known, drawn independently for each country, and are independent of θ .

There are two information sources that countries have at their disposal when determining whether to implement climate policy. The first is the history of observed actions, $h_i = (a_1, \dots, a_{i-1})$. Countries learn about the suitability of green policy θ through the behavior of others. They also receive conditionally independent private signals, which, along with the prior, generate private beliefs $p_i \in [0, 1]$. I work with these beliefs rather than signals and the prior π directly (p_i is a sufficient statistic). These beliefs are not publicly known, but insofar as they translate into actions they may partially be inferred. Let the cumulative distribution function of a private belief p in state θ be $F(p|\theta)$ with density $f(p|\theta)$ such that $F(p|1) < F(p|0)$.⁶

⁶By Bayes's Rule, the state-conditional densities $f(p|\theta)$ satisfy $p = \frac{\pi f(p|1)}{f(p)}$ and $1 - p = \frac{(1-\pi)f(p|0)}{f(p)}$ with $f(p) = \pi f(p|1) + (1-\pi)f(p|0)$. Then $\frac{f(p|1)}{f(p|0)} = \frac{p}{1-p} \frac{1-\pi}{\pi}$; this stochastic ordering implies that the conditional distributions are mutually absolutely continuous, share the same support, and that $F(p|1) < F(p|0)$ for all private beliefs strictly inside the support.

Given some history of countries' climate policies h_i , define the public belief $P(\theta = 1|h_i)$ as the informational content about the suitability of the green transition. There is an associated public likelihood ratio $\ell_i = \frac{1-P(\theta=1|h_i)}{P(\theta=1|h_i)}$ such that lower values of ℓ_i imply a greater likelihood that $\theta = 1$, or that a green transition would be successful.

The climate problem is often described as one of strategic substitutes because it is nationally costly to exert effort to address climate change despite that this effort provides a global benefit. To capture this tension, I introduce *collective action penalties*, which are action-specific, history-dependent costs (Eyster et al. 2014). It becomes more costly to pursue climate reform if many other countries have done so already. In reduced form, these penalties capture the strategic substitutability of climate actions across countries present in other models. Denote collective actions penalties as $z(h_i)$ (and suppress dependence on h_i where it is not confusing). This function is increasing in the number of countries who have already taken climate action, or $\sum h_i$. For simplicity, assume the penalty is bounded, $z_i \in [0, 1]$; since the domestic costs c_i are on the same scale, there is no explicit assumption as to whether implementation costs are greater than collective action penalties. As an example, consider a linearly proportional cost function for any country $i \geq 2$ (with $z_1 = 0$),

$$z_i = \frac{\sum h_i}{i-1}.$$

In countries' payoffs I scale these externalities by $k > 0$ in order to parameterize the extent to which countries weigh potential complementarities (generated by information about θ) and potential substitutes (generated by collective action penalties). This parameter can be thought of as scaling the extent to which countries internalize free-riding concerns; larger k implies stronger free-riding incentives as collective action penalties are weighted more heavily.

Given this setup, country i 's payoff can be written as

$$u_i(a_i, h_i, \theta; c_i) = a_i(\theta - c_i - kz(h_i)).$$

A strategy for country i is a choice to implement climate reforms or not, $a_i \in \{0, 1\}$, given the choices of other prior-moving countries contained in history h_i , and its type (p_i, c_i) , comprised of its private belief about θ and its domestic costs of implementing green policy. I examine weak perfect Bayesian equilibria and derive all posterior beliefs via Bayes's Rule.

F.2 Results and Proofs

Fix a history $h_i = (a_1, \dots, a_{i-1})$ of past climate policy adoption decisions that induce a public likelihood ratio ℓ_i and potential collective action penalties z_i . Given ℓ_i and the private belief p_i , country i can make an assessment about the appropriateness of climate policy, i 's posterior belief that $\theta = 1$ is defined as

$$\mu_i = P(\theta = 1|p_i, \ell_i) = \frac{p_i}{p_i + (1 - p_i)\ell_i}.$$

Then, country i prefers to implement climate reforms if and only if

$$\begin{aligned}
\mu_i - c_i - kz_i &\geq 0 \\
\Leftrightarrow \mu_i &\geq c_i + kz_i \\
\Leftrightarrow p_i &\geq \frac{\ell_i(c_i + kz_i)}{1 - (c_i + kz_i) + \ell_i(c_i + kz_i)} \equiv \tilde{p}(c_i, \ell_i).
\end{aligned}$$

Country i pursues climate action if and only if their private belief about a successful green transition is sufficiently high, given the domestic political costs of implementing climate reforms and the potential collective action penalties. If $\tilde{p}(c_i, \ell_i) > 1$, then i never takes climate action regardless of the value of p_i , which occurs whenever $c_i > 1 - kz_i \equiv \bar{c}_i$. Intuitively, if the domestic costs of implementing climate policy are prohibitively high, it does not matter how optimistic i is about the green transition, implementing green policy is not domestically feasible. Then, for any $c_i \in [0, \bar{c}_i]$, i pursues climate action if and only if $p_i > \tilde{p}(c_i, \ell_i)$, which occurs with probability $1 - F(\tilde{p}(c_i, \ell_i)|\theta)$.

Lemma A.1. *The threshold $\tilde{p}(c_i, \ell_i)$ is:*

- *increasing in the public likelihood ratio ℓ_i ;*
- *increasing in domestic implementation costs c_i ;*
- *increasing in collective action penalties z_i ;*
- *increasing in the strength of free-riding incentives k .*

Proof of Lemma A.1.

$$\begin{aligned}
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial \ell_i} &= \frac{(1 - c_i - kz_i)(c_i + kz_i)}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0. \\
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial c_i} &= \frac{\ell_i}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0. \\
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} &= \frac{k\ell_i}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0. \\
\frac{\partial \tilde{p}(c_i, \ell_i)}{\partial k} &= \frac{\ell_i z_i}{(1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2} \geq 0.
\end{aligned}$$

□

Proposition A.1. *Let $\alpha^*(a_i|\ell_i, \theta)$ be the probability that country i takes climate action a_i in state θ . Then*

$$\alpha^*(1|\ell_i, \theta) = \int_0^{\bar{c}_i} 1 - F(\tilde{p}(c_i, \ell_i)|\theta) dc_i = 1 - \alpha^*(0|\ell_i, \theta).$$

Proof of Proposition A.1. Immediate from text. $\square$

Corollary A.1. *Climate policy is informative about θ . The probability of climate action is greater when $\theta = 1$ versus $\theta = 0$: $\alpha^*(1|\ell_i, 1) > \alpha^*(1|\ell_i, 0)$. The probability of climate inaction is greater when $\theta = 0$ versus $\theta = 1$: $\alpha^*(0|\ell_i, 1) < \alpha^*(0|\ell_i, 0)$.*

Proof of Corollary A.1.

$$\begin{aligned}\alpha^*(1|\ell_i, 1) - \alpha^*(1|\ell_i, 0) &= \left(\int_0^{\bar{c}_i} 1 - F(\tilde{p}(c_i, \ell_i)|1) \, dc_i \right) - \left(\int_0^{\bar{c}_i} 1 - F(\tilde{p}(c_i, \ell_i)|0) \, dc_i \right) \\ &= \int_0^{\bar{c}_i} F(\tilde{p}(c_i, \ell_i)|0) - F(\tilde{p}(c_i, \ell_i)|1) \, dc_i > 0,\end{aligned}$$

where the result follows from the stochastic ordering of private beliefs. $\square$

Corollary A.2. *The probability of climate action is increasing in the public optimism about a successful green transition, $\frac{d\alpha^*(1|\ell_i, \theta)}{d\ell_i} \leq 0$.*

Proof of Corollary A.2. Differentiating with respect to ℓ_i yields

$$\frac{d\alpha^*(1|\ell_i, \theta)}{d\ell_i} = - \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}_i}{\partial \ell_i} \, dc_i \leq 0.$$

$\square$

Corollary A.3. *The probability of climate action is decreasing in collective action penalties, $\frac{d\alpha^*(1|\ell_i, \theta)}{dz_i} \leq 0$.*

Proof of Corollary A.3. By the Leibniz integral rule, differentiating with respect to z_i yields

$$\begin{aligned}\frac{d\alpha^*(1|\ell_i, \theta)}{dz_i} &= \frac{\partial \bar{c}_i}{\partial z_i} - F(\tilde{p}(\bar{c}_i, \ell_i)|\theta) \frac{\partial \bar{c}_i}{\partial z_i} - \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} \, dc_i \\ &= - \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} \, dc_i \leq 0.\end{aligned}$$

where the first two terms simplify because $\tilde{p}(\bar{c}_i, \ell_i) = 1$. $\square$

Corollary A.2 states that more optimistic public beliefs about a successful green transition begets more climate action. That is, these beliefs endogenously generate *complementarities* in countries' climate actions. Conversely, increased collective action penalties—which arise because more countries have already pursued climate policies—depress subsequent climate action, as stated in Corollary A.3. The actions of prior movers induce *substitution* in the behavior of later policymakers.

Which factor dominates? Under what conditions are countries' actions strategic complements or strategic substitutes in equilibrium? To conceptualize this, I consider the ratio of the marginal effects of public beliefs and collective action penalties. Define $\rho(\ell_i, z_i|\theta)$ as

$$\rho(\ell_i, z_i|\theta) = \frac{d\alpha^*(1|\ell_i, \theta)}{d\ell_i} \bigg/ \frac{d\alpha^*(1|\ell_i, \theta)}{dz_i}.$$

The magnitude of $\rho(\ell_i, z_i|\theta)$ is always positive, but we can think about which factor dominates—strategic complementarities that stem from increased public beliefs or strategic substitutes from collective action penalties—based on where it is greater than or less than 1. If $\rho(\ell_i, z_i|\theta) > 1$, then, all else equal, varying public beliefs has a larger effect on the equilibrium probability of climate action than does varying collective action penalties. In this case, we can say that the net effect of other countries' behavior generates complementarities for country i . By contrast, when $\rho(\ell_i, z_i|\theta) < 1$, then the incentives to free ride swamp the potential benefits from climate policy investment.

Proposition A.2. *Complementarity effects dominate when free-riding incentives are small, and substitution effects dominate when free-riding incentives are large: there exists a threshold $\bar{k}_i$ such that if $k < \bar{k}_i$ then $\rho(\ell_i, z_i|\theta) > 1$.*

Proof of Proposition A.2. It follows that

$$\rho(\ell_i, z_i|\theta) > 1 \Leftrightarrow \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}_i}{\partial \ell_i} dc_i > \int_0^{\bar{c}_i} f(\tilde{p}(c_i, \ell_i)|\theta) \frac{\partial \tilde{p}(c_i, \ell_i)}{\partial z_i} dc_i.$$

Define $Q(k) = (1 - (c_i + kz_i) + \ell_i(c_i + kz_i))^2 \geq 0$, which is the denominator of the comparative statics on $\tilde{p}(c_i, \ell_i)$. Simplifying yields

$$\int_0^{\bar{c}_i} \frac{f(\tilde{p}(c_i, \ell_i)|\theta)}{Q(k)} \left((1 - c_i - k_i)(c_i + kz_i) - k\ell_i \right) dc_i > 0.$$

Now note that for any $k < \frac{1}{z_i}$, the integral is well-defined (otherwise $\bar{c}_i = 0$). Furthermore, for any $k < \frac{1}{z_i}$, $\frac{f(\tilde{p}(c_i, \ell_i)|\theta)}{Q(k)} \geq 0$ and we are integrating over a positive interval of the c_i space. So the integrand is negative if and only if

$$(1 - c_i - k_i)(c_i + kz_i) - k\ell_i < 0,$$

which simplifies to $k > \frac{z_i - 2c_i z_i - \ell_i + \sqrt{\ell_i^2 - 2\ell_i z_i + 4c_i \ell_i z_i + z_i^2}}{2z_i^2} \equiv \bar{k}_i$. Hence a sufficient condition for the integrand to be negative is if $k > \bar{k}_i$ which implies that $\rho(\ell_i, z_i|\theta) < 1$. $\square$

We can now use the model to think about the long-run dynamics of climate policy across countries based on the analysis in the previous subsection. We have shown that the decision

problem facing each country at the time of climate adoption is static, meaning history-relevant parameters such as ℓ_i and z_i can be treated in reduced form, but now wish to trace the evolution of actions and beliefs across countries.

Since private signals are conditionally independent, the likelihood ratio updates such that

$$\ell_{i+1} = \varphi(a_i, \ell_i) = \ell_i \frac{\alpha^*(a_i|\ell_i, 0)}{\alpha^*(a_i|\ell_i, 1)}.$$

Observe that by Corollary A.1, relative to ℓ_i , ℓ_{i+1} shrinks if $a_i = 1$ but ℓ_{i+1} grows if $a_i = 0$. The public belief becomes more or less optimistic depending on the previous action a_i , which in turn informs the decision to enact climate policy in the subsequent period. Then, given the updated public belief and any additional collective action penalties, country $i+1$ considers the tradeoff between implementing climate reforms and incurring domestic implementation costs and collective action penalties or free-riding, where climate policy occurs with probability $\alpha^*(1|\ell_{i+1}, \theta)$.

As is standard in the informational cascades and herding literature (e.g., [Smith and Sørensen 2000](#)), convergence results are stated conditioning on $\theta = 1$. This is also the more interesting case from a substantive perspective anyway, as this is where the tradeoff between the two mechanisms is present.

Lemma A.2. *Conditional on $\theta = 1$, the public likelihood ratio $\langle \ell_i \rangle$ is a martingale.*

Proof of Lemma A.2. Recall that the public likelihood ratio updates according to

$$\ell_{i+1} = \ell_i \frac{\alpha^*(a_i|\ell_i, 0)}{\alpha^*(a_i|\ell_i, 1)},$$

by the conditional independence of signals. Taking expectations yields

$$\begin{aligned} E[\ell_{i+1}|\ell_1, \dots, \ell_i, \theta = 1] &= \sum_{a \in \{0,1\}} \alpha^*(a|\ell_i, 1) \ell_i \frac{\alpha^*(a|\ell_i, 0)}{\alpha^*(a|\ell_i, 1)} \\ &= \ell_i \sum_{a \in \{0,1\}} \alpha^*(a|\ell_i, 0) \\ &= \ell_i. \end{aligned}$$

□

Proposition A.3. *In the limit, countries learn whether the green transition will be successful: public beliefs converge to the true state of the world almost surely.*

Proof of Proposition A.3. Without loss of generality condition on state $\theta = 1$. Since $\langle \ell_i \rangle$ is a martingale and all values are nonnegative, it converges almost surely to a random variable $\ell_\infty = \lim_{i \rightarrow \infty} \ell_i$ with support $[0, \infty)$ by the Martingale Convergence Theorem ([Doob 1953](#)).

This rules out nonstationary limit beliefs. Since private beliefs p_i are unbounded within $[0, 1]$, then the only stationary finite likelihood ratio in state $\theta = 1$ is 0, so $\ell_\infty \rightarrow 0$ almost surely (Smith and Sørensen 2000, Theorem 1). $\square$

Proposition A.4. *In the limit, countries take the correct action $a_i = \theta$ if and only if $c_i \leq \bar{c}_i$.*

Proof of Proposition A.4. Recall that i chooses $a_i = 0$ if $c_i > \bar{c}_i$ for any private belief p_i , which occurs with probability $P(c_i > \bar{c}_i) = kz_i$. This probability is increasing in z_i , which increases in the number of countries that choose $a_i = 1$. Now suppose that $\theta = 1$ and $c_i \leq \bar{c}_i$ so i chooses $a_i = 1$ iff $p_i \geq \tilde{p}(c_i, \ell_i)$. By Proposition A.3, the likelihood ratio converges almost surely to $\ell_i \rightarrow 0$. Then we have $\lim_{\ell_i \rightarrow 0} \tilde{p}(c_i, \ell_i) = 0$. Hence conditional on $c_i < \bar{c}_i$, i chooses $a_i = 1 = \theta$ for any private belief.

If countries are taking action on a measure zero subset, $\bar{c}_j \rightarrow 0$ or $k > \frac{1}{z_j}$ for some $j \leq n$, then countries pool on $a_j = 0 \forall j, \dots, n$. $\square$

Corollary A.4. *Let $z_i(h_i) = \frac{\sum_{i=1}^n h_i}{i-1}$. The probability of climate action converges to $\frac{1}{1+k}$.*

Proof of Corollary A.4. Conditional on state $\theta = 1$, climate action occurs with probability $\bar{c}_i$, as $\tilde{p}(c_i, \ell_i) \rightarrow 0$ as $\ell_i \rightarrow 0$. Moreover, given that $z_i = \frac{\sum_{i=1}^n h_i}{i-1}$, it is a linear proportional function of previous actions, so in the limit, $z_i \rightarrow \alpha^*(1|\ell_i, \theta)$. Then we have

$$\begin{aligned} \alpha^*(1|\ell_i, \theta) &= \bar{c}_i = 1 - kz_i \\ &= 1 - k\alpha^*(1|\ell_i, \theta) \\ \Leftrightarrow \alpha^*(1|\ell_i, \theta) &= \frac{1}{1+k}. \end{aligned}$$

$\square$

Corollary A.5. *In the limit, $\rho(\ell_i, z_i|\theta) > 1$ if $k < 1 - c_i$.*

Proof of Corollary A.5. Per Proposition A.2, a sufficient condition for $\rho(\ell_i, z_i|\theta) > 1$ is $k < \bar{k}_i$ where $\bar{k}_i = \frac{z_i - 2c_i z_i - \ell_i + \sqrt{\ell_i^2 - 2\ell_i z_i + 4c_i \ell_i z_i + z_i^2}}{2z_i^2}$. Then in the long run, $\lim_{\ell_i \rightarrow 0} \bar{k}_i = 1 - c_i$. $\square$

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